

Assessing PVA Predictions

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Background

[Hooker et al. \(2020\)](#) used noninvasive sampling and spatial capture-recapture models to estimate the abundance of female black bears in central Georgia. Their analysis yielded point estimates of N_t between 106 and 166 using samples collected between 2012 and 2016. Using the same model, Hooker et al. (2020) also conducted population viability analyses (PVA) for the central GA black bear population. They projected abundance 50 years into the future using survival and density-dependent recruitment rates under management scenarios where a Poisson-distributed number of additional females (above current levels) were removed from the population by harvest. They evaluated *status quo* simulations (no additional harvest) and four alternatives where an average of $h = 5, 10, 15$, or 20 females (above *status quo* levels) were removed from the population by harvest. They found that 50-year extinction risk remained $<1\%$ in both the *status quo* and $h = 5$ simulations (see Figure 6 of [Hooker et al. 2020](#)).

In this markdown, we will revisit predictions from these simulations and use data for *known* female mortalities (which likely underestimates total mortality) to assess i) which scenario considered in the PVA from Hooker et al. (2020) is most similar to the recent past in central GA? and ii) which scenario do estimates of abundance from the present study (see `13_fitHooker_bothSexes`) align with most closely?

For the latter comparison, we'll use estimates from the single γ model, which estimates more realistic increases in female abundance between 2023 and 2024.

This markdown will cover the following analyses:

- i. Characterize observed mortality levels in central Georgia (for comparison to alternative scenarios considered in Hooker et al. (2020))
 - ii. Refit the Hooker et al. (2020) model to the capture history from that study (using scripts and data from the project's [GitHub repository](#))
 - iii. Repeat the PVA forecasting simulations from Hooker et al. (2020) with $h = 0, 5, 10, 15$, and 20 (as in the original paper), again using scripts from their study
 - iv. Compare abundance forecasts from scenarios considered in Hooker et al. (2020) to posterior distributions of female abundance from the SCR model fit in this study ("v1" model, no sex-specific recruitment)
-

Observed mortality in central Georgia

The Georgia Department of Natural Resources retains records of central GA black bear mortality from various sources (*e.g.*, legal harvest, illegal take, vehicular collisions, etc.). We will use those records to:

- i. Visualize trends in mortality from these sources over time
- ii. Calculate average numbers of deaths for three time windows
 - pre-2012: before the Hooker et al. (2020) study period
 - 2012-2016: the Hooker et al. (2020) study period, corresponding most closely to rates in their PVA
 - 2017-2024: prior to, and including, the current study period
- iii. Assess predictions from the PVA in Hooker et al. (2020), given female abundance estimates from our model (`13_fitHooker_bothSexes`) and observed female mortality data

Dataset

First load the dataset. The `.csv` file in the repository transposes data from [this spreadsheet](#) (see the **CGA Trends** tab). Below, we read the data into R, and create subsets for the two sexes and the different sources of mortality. Note that data from 2025 were not complete at the time this spreadsheet was created, so we focus on the period between 2003 and 2024.

```
mort <- read.csv("../input/CGAmortalities.csv", header = TRUE)
mort
```

##	year	harvestM	harvestF	harvestU	illegalM	illegalF	illegalU	roadM	roadF	roadU
## 1	2003	0	0	0	0	0	0	0	2	0
## 2	2004	0	0	0	0	1	0	1	1	0
## 3	2005	0	0	0	0	0	0	8	3	0
## 4	2006	0	0	0	1	0	0	4	1	0
## 5	2007	0	0	0	0	0	0	5	1	1
## 6	2008	0	0	0	0	0	1	5	1	3
## 7	2009	1	0	0	0	1	0	7	3	1
## 8	2010	0	2	0	0	2	0	3	1	1
## 9	2011	15	17	0	2	0	0	2	1	2
## 10	2012	4	7	0	2	2	0	5	2	0
## 11	2013	0	1	0	0	0	0	4	1	0
## 12	2014	1	3	0	0	0	0	7	1	0
## 13	2015	3	9	0	2	1	0	7	2	1

```
## 14 2016      5      5      0      3      0      0      6      4      1
## 15 2017      0      0      0      1      0      0     13      3      0
## 16 2018      1      1      0      1      0      0     14      4      2
## 17 2019      1      0      0      1      1      0      6      3      2
## 18 2020      2      3      0      0      3      0      8      1      1
## 19 2021      1      5      0      0      1      0     12      4      3
## 20 2022      4     10      0      1      0      0      5      1      0
## 21 2023      2      1      0      0      3      0     12      6      1
## 22 2024      2      3      0      0      4      0      8      2      1
## 23 2025      0      0      0      0      0      0      0      1      0
```

```
#2025 was a partial year in this dataset, remove it
mort <- mort[-which(mort$year == 2025),]

#total mortality by sex
total.male <- rowSums(mort[,which(colnames(mort) %in% c("harvestM",
                                                       "illegalM",
                                                       "roadM"))])

total.female <- rowSums(mort[,which(colnames(mort) %in% c("harvestF",
                                                         "illegalF",
                                                         "roadF"))])

#Combined
all.mort <- data.frame(year = mort$year,
                      harvest = mort$harvestM+mort$harvestF+mort$harvestU,
                      illegal = mort$illegalM+mort$illegalF+mort$illegalU,
                      road = mort$roadM+mort$roadF+mort$roadU)
(total.all <- rowSums(all.mort[, -1]))
```

```
## [1] 2 3 11 6 7 10 13 9 39 22 6 12 25 24 17 23 14 18 26 21 25 20
```

Visualize trends

Now to visualize trends through time in the different sources.

Both sexes

First, for all black bears in central GA.

```
all.mort

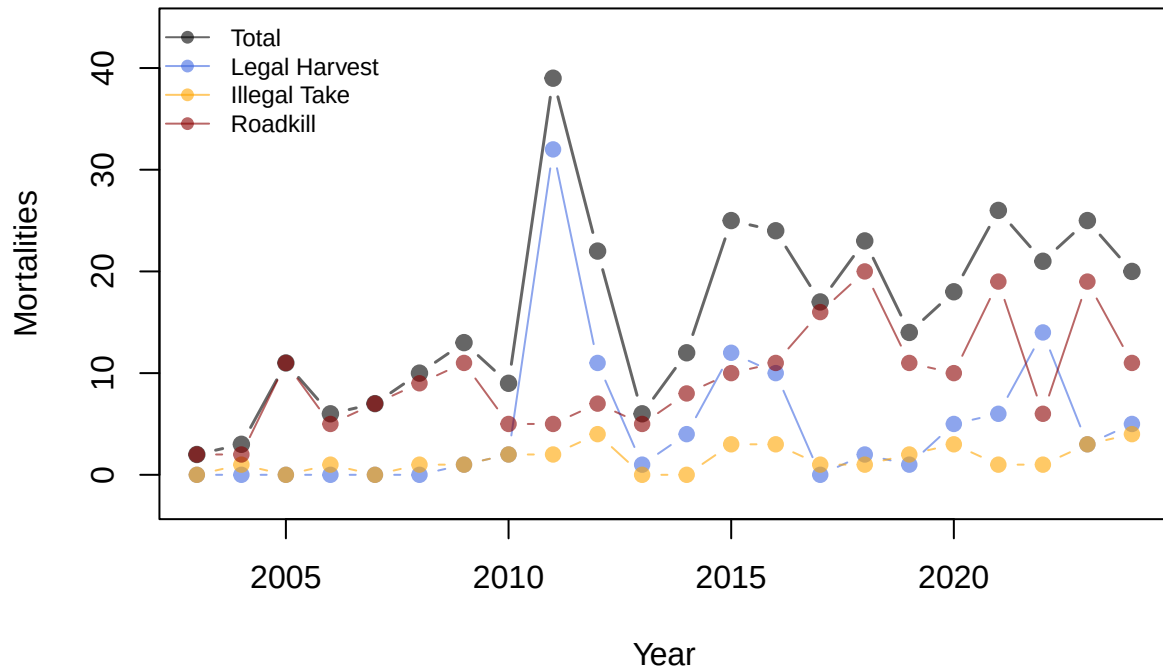
##   year harvest illegal road
## 1 2003      0       0    2
## 2 2004      0       1    2
## 3 2005      0       0   11
## 4 2006      0       1    5
## 5 2007      0       0    7
## 6 2008      0       1    9
## 7 2009      1       1   11
## 8 2010      2       2    5
## 9 2011     32       2    5
## 10 2012     11       4    7
```

```
## 11 2013      1      0      5
## 12 2014      4      0      8
## 13 2015     12      3     10
## 14 2016     10      3     11
## 15 2017      0      1     16
## 16 2018      2      1     20
## 17 2019      1      2     11
## 18 2020      5      3     10
## 19 2021      6      1     19
## 20 2022     14      1      6
## 21 2023      3      3     19
## 22 2024      5      4     11
```

```
library(scales)
color.pal <- c(alpha("black", 0.6),
               alpha("royalblue", 0.6),
               alpha("orange", 0.6),
               alpha("darkred",0.6))

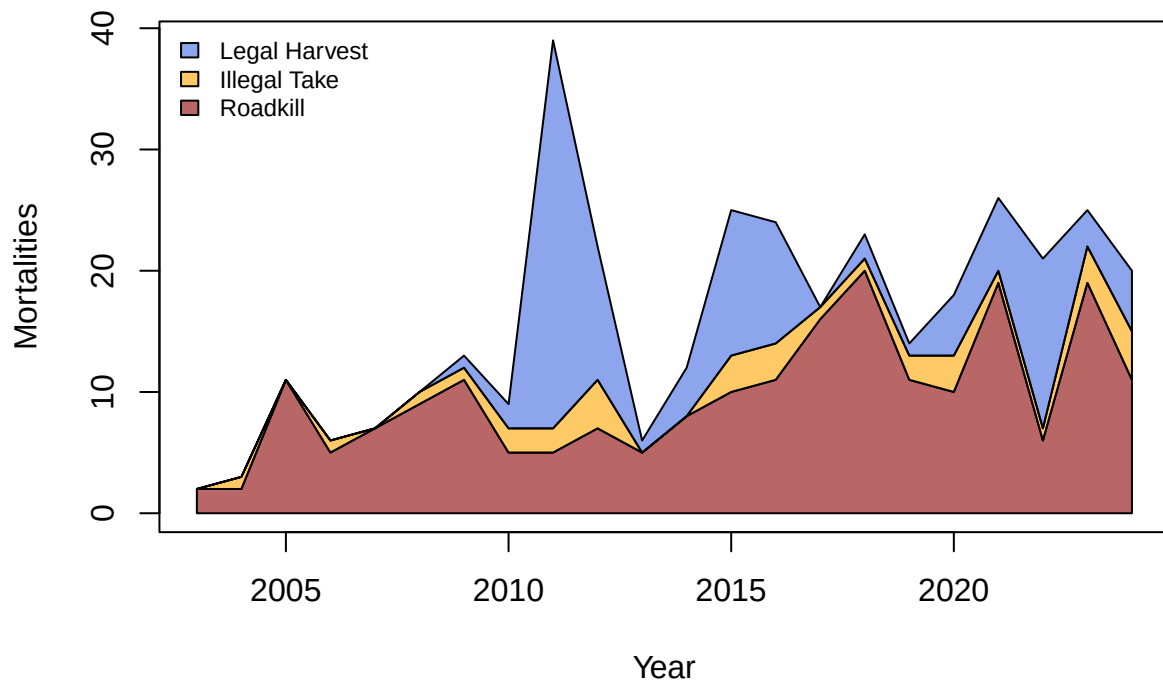
#As a scatter plot
plot(total.all ~ all.mort$year,
     type = "b", lwd = 1.5, pch=19, col = color.pal[1],
     ylim = c(-2.5,(max(total.all)+5)),
     xlab = "Year", ylab = "Mortalities", main = "All Bears")
lines(all.mort$harvest ~ all.mort$year,
      pch=19, type = "b", col = color.pal[2])
lines(all.mort$illegal ~ all.mort$year,
      pch=19, type = "b", col = color.pal[3])
lines(all.mort$road ~ all.mort$year,
      pch=19, type = "b", col = color.pal[4])
legend("topleft", cex = 0.75, bty="n",
      lty=1, pch=19, col=color.pal,
      legend=c("Total","Legal Harvest","Illegal Take","Roadkill"))
```

All Bears

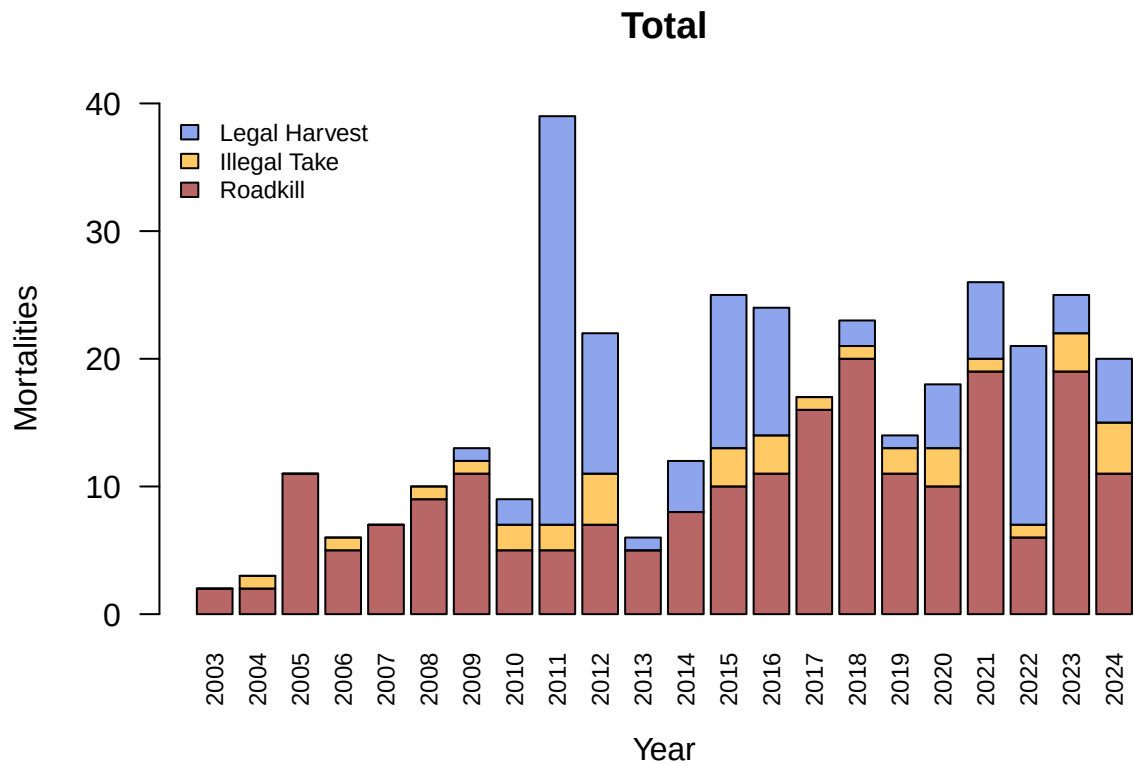


```
#As an area chart
library(areaplot)
areaplot(x=mort$year,
         y=all.mort[,-1],
         col=color.pal[2:4],
         xlab = "Year", ylab = "Mortalities", main = "All Bears")
legend("topleft", cex = 0.75, bty="n",
      fill=color.pal[2:4],
      legend=c("Legal Harvest","Illegal Take","Roadkill"))
```

All Bears



```
#As a stacked barplot
barplot(height = rbind(all.mort$road, all.mort$illegal, all.mort$harvest),
        beside = FALSE, names.arg = all.mort$year, col = color.pal[4:2],
        xlab = "Year", ylab = "Mortalities", main = "Total",
        ylim = c(0,40), las = 2, cex.names = 0.75)
legend("topleft", cex = 0.75, bty="n",
       fill=color.pal[2:4],
       legend=c("Legal Harvest","Illegal Take","Roadkill"))
```



Females

Now for females only...

```
mort[,which(colnames(mort) %in% c("year","harvestF","illegalF","roadF"))]
```

```
##   year harvestF illegalF roadF
## 1  2003         0         0     2
## 2  2004         0         1     1
## 3  2005         0         0     3
## 4  2006         0         0     1
## 5  2007         0         0     1
## 6  2008         0         0     1
## 7  2009         0         1     3
## 8  2010         2         2     1
## 9  2011        17         0     1
##10  2012         7         2     2
##11  2013         1         0     1
##12  2014         3         0     1
##13  2015         9         1     2
##14  2016         5         0     4
##15  2017         0         0     3
##16  2018         1         0     4
##17  2019         0         1     3
##18  2020         3         3     1
```

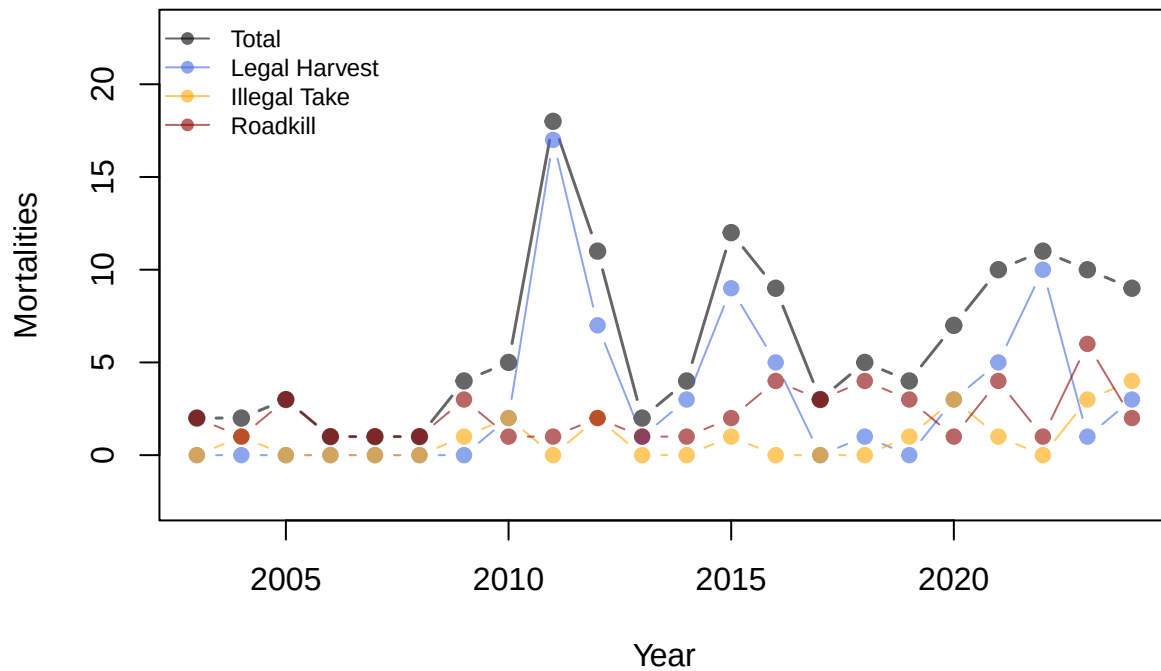
```
## 19 2021      5      1      4
## 20 2022     10      0      1
## 21 2023      1      3      6
## 22 2024      3      4      2
```

```
mean(total.female)
```

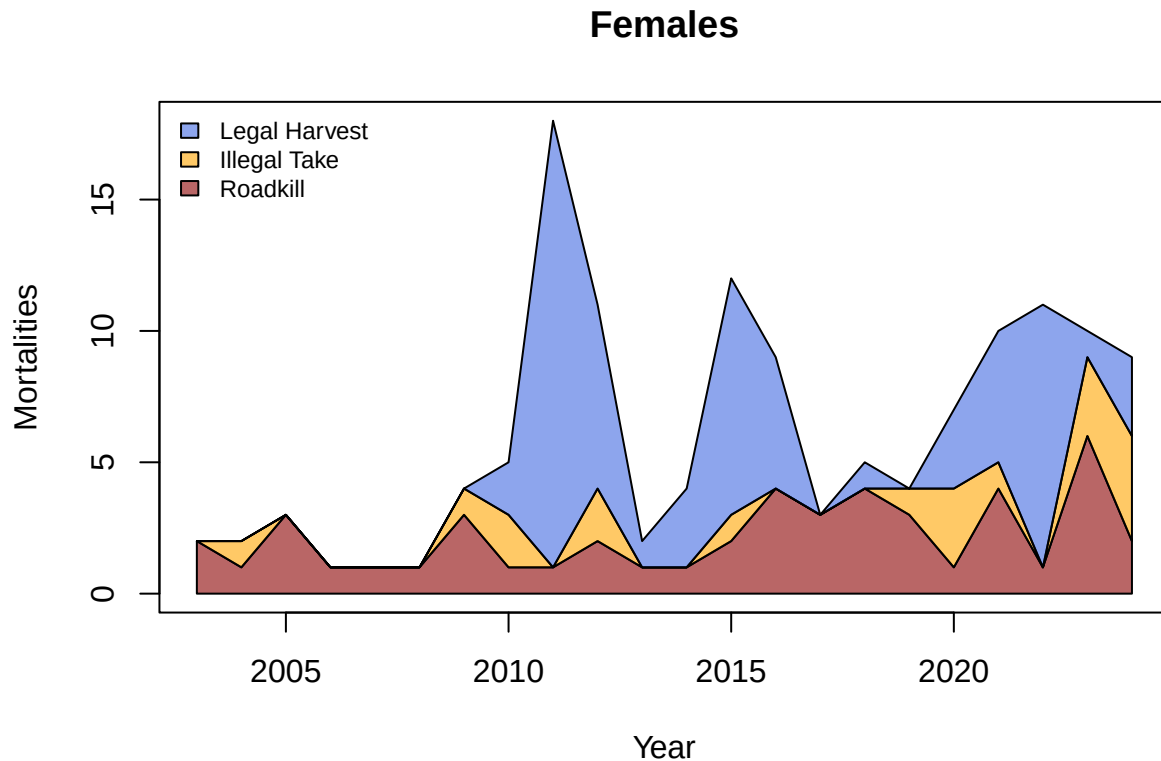
```
## [1] 6.090909
```

```
#Scatter plot
plot(total.female ~ mort$year,
     type = "b", lwd = 1.5, pch=19, col = color.pal[1],
     ylim = c(-2.5,(max(total.female)+5)),
     xlab = "Year", ylab = "Mortalities", main = "Females")
lines(mort$harvestF ~ mort$year,
     pch=19, type = "b", col = color.pal[2])
lines(mort$illegalF ~ mort$year,
     pch=19, type = "b", col = color.pal[3])
lines(mort$roadF ~ mort$year,
     pch=19, type = "b", col = color.pal[4])
legend("topleft", cex = 0.75, bty="n",
     lty=1, pch=19, col=color.pal,
     legend=c("Total","Legal Harvest","Illegal Take","Roadkill"))
```

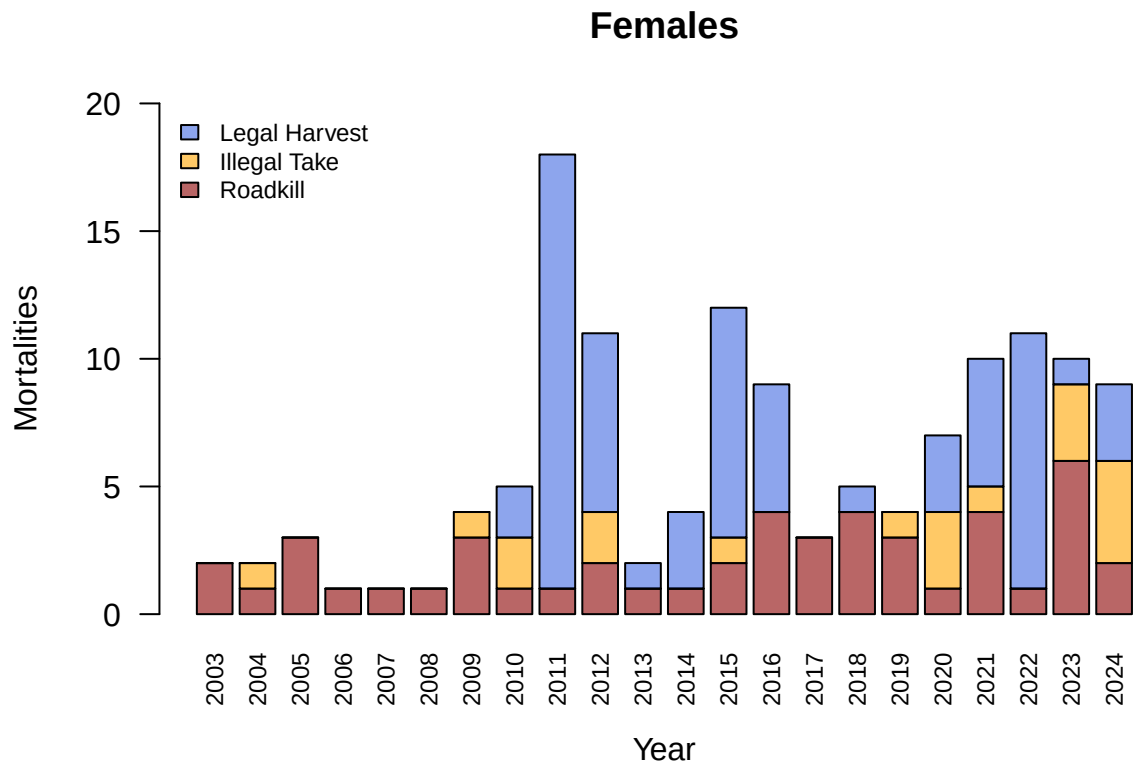
Females




```
#Area chart
areaplot(x=mort$year,
         y=cbind(mort$harvestF, mort$illegalF, mort$roadF),
         col=color.pal[2:4],
         xlab = "Year", ylab = "Mortalities", main = "Females")
legend("topleft", cex = 0.75, bty="n",
       fill=color.pal[2:4],
       legend=c("Legal Harvest","Illegal Take","Roadkill"))
```



```
#Stacked barplot
barplot(height = rbind(mort$roadF, mort$illegalF, mort$harvestF),
        beside = FALSE, names.arg = mort$year, col = color.pal[4:2],
        xlab = "Year", ylab = "Mortalities", main = "Females",
        ylim = c(0,20), las = 2, cex.names = 0.75)
legend("topleft", cex = 0.75, bty="n",
       fill=color.pal[2:4],
       legend=c("Legal Harvest","Illegal Take","Roadkill"))
```



Notice from this plot that vehicular collisions are less frequent in females.

Males

```
mort[,which(colnames(mort) %in% c("year", "harvestM", "illegalM", "roadM"))]
```

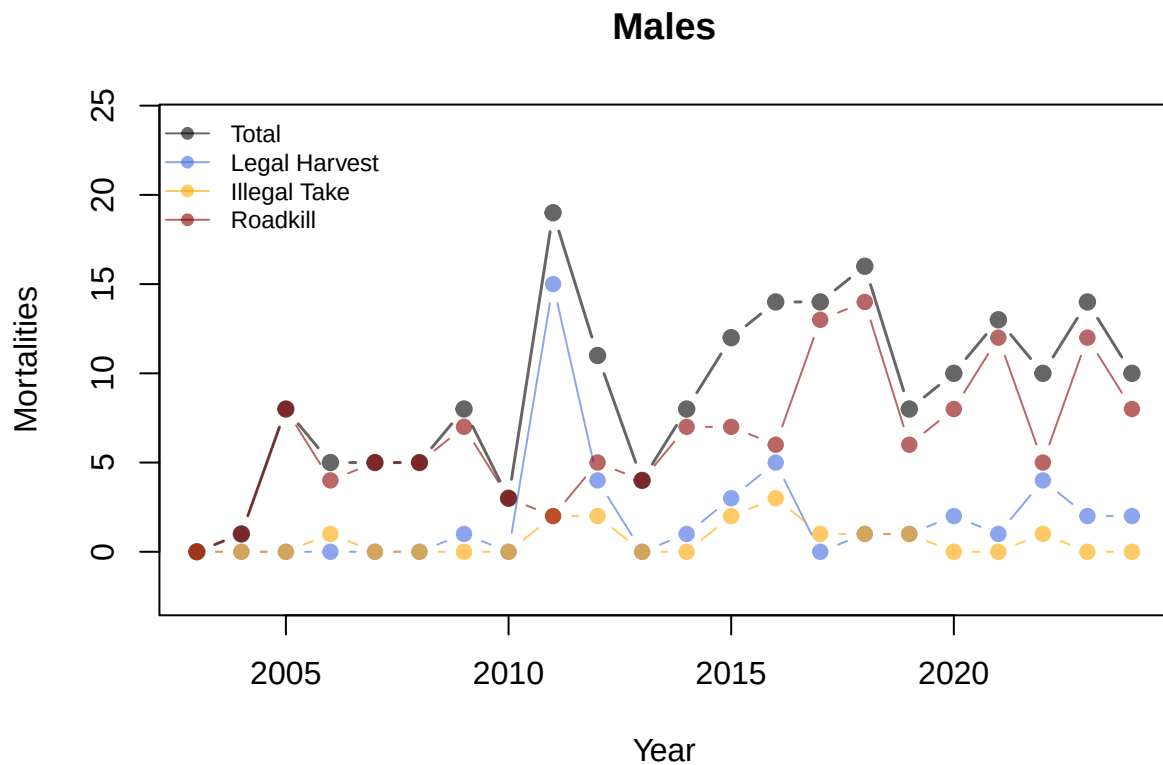
```
##   year harvestM illegalM roadM
## 1  2003         0         0     0
## 2  2004         0         0     1
## 3  2005         0         0     8
## 4  2006         0         1     4
## 5  2007         0         0     5
## 6  2008         0         0     5
## 7  2009         1         0     7
## 8  2010         0         0     3
## 9  2011        15         2     2
##10  2012         4         2     5
##11  2013         0         0     4
##12  2014         1         0     7
##13  2015         3         2     7
##14  2016         5         3     6
##15  2017         0         1    13
##16  2018         1         1    14
##17  2019         1         1     6
```

```
## 18 2020      2      0      8
## 19 2021      1      0     12
## 20 2022      4      1      5
## 21 2023      2      0     12
## 22 2024      2      0      8
```

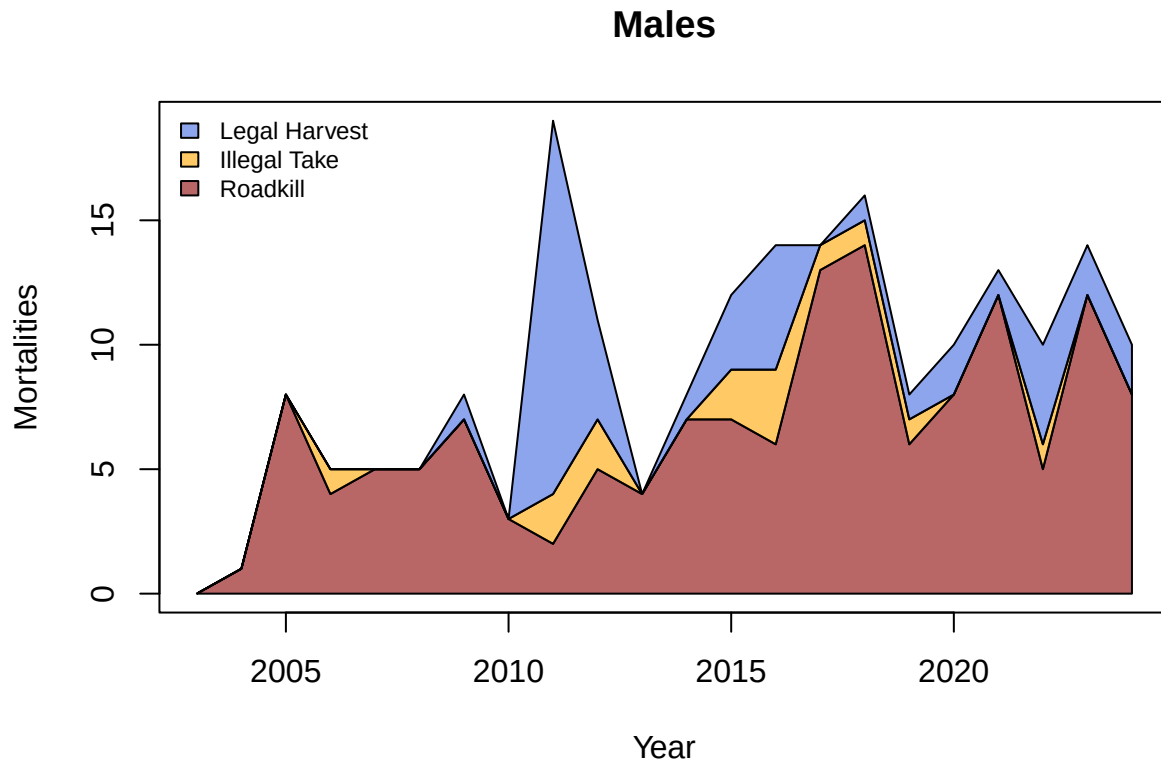
```
mean(total.male)
```

```
## [1] 9
```

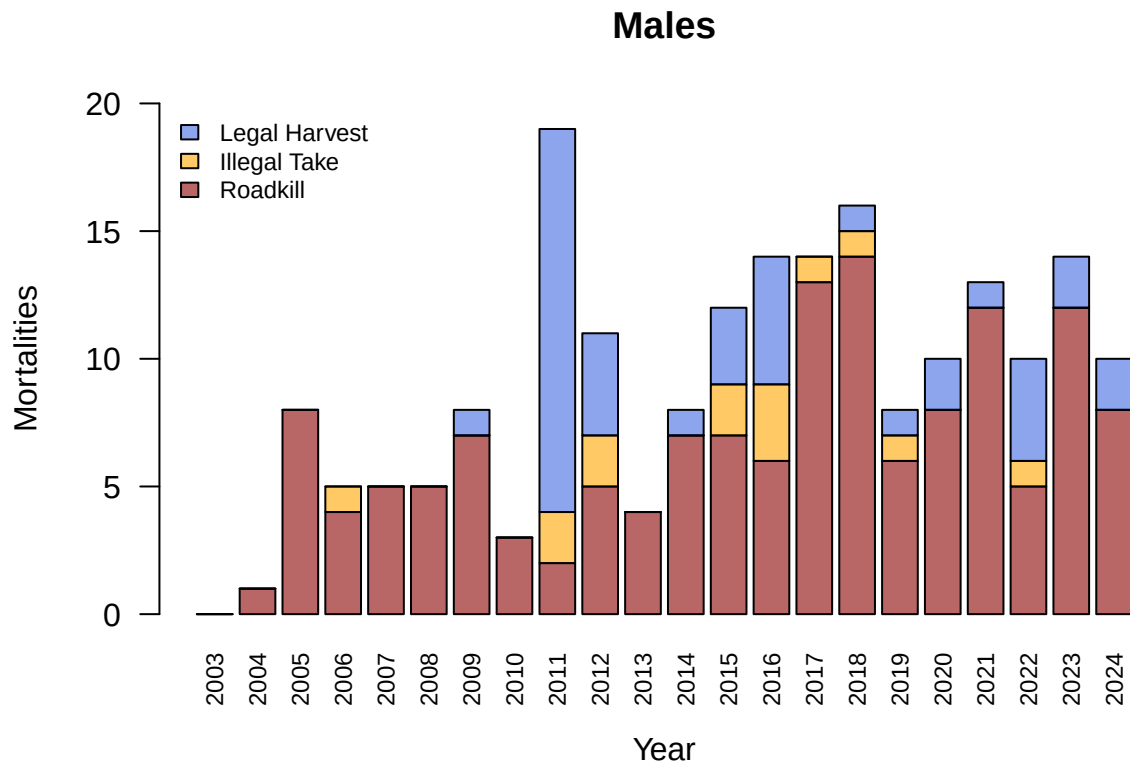
```
#Scatter plot
plot(total.male ~ mort$year,
     type = "b", lwd = 1.5, pch=19, col = color.pal[1],
     ylim = c(-2.5,(max(total.male)+5)),
     xlab = "Year", ylab = "Mortalities", main = "Males")
lines(mort$harvestM ~ mort$year,
     pch=19, type = "b", col = color.pal[2])
lines(mort$illegalM ~ mort$year,
     pch=19, type = "b", col = color.pal[3])
lines(mort$roadM ~ mort$year,
     pch=19, type = "b", col = color.pal[4])
legend("topleft", cex = 0.75, bty="n",
     lty=1, pch=19, col=color.pal,
     legend=c("Total","Legal Harvest","Illegal Take","Roadkill"))
```



```
#Area chart
areaplot(x=mort$year,
         y=cbind(mort$harvestM, mort$illegalM, mort$roadM),
         col=color.pal[2:4],
         xlab = "Year", ylab = "Mortalities", main = "Males")
legend("topleft", cex = 0.75, bty="n",
       fill=color.pal[2:4],
       legend=c("Legal Harvest", "Illegal Take", "Roadkill"))
```



```
#Stacked barplot
barplot(height = rbind(mort$roadM, mort$illegalM, mort$harvestM),
        beside = FALSE, names.arg = mort$year, col = color.pal[4:2],
        xlab = "Year", ylab = "Mortalities", main = "Males",
        ylim = c(0,20), las = 2, cex.names = 0.75)
legend("topleft", cex = 0.75, bty="n",
       fill=color.pal[2:4],
       legend=c("Legal Harvest", "Illegal Take", "Roadkill"))
```



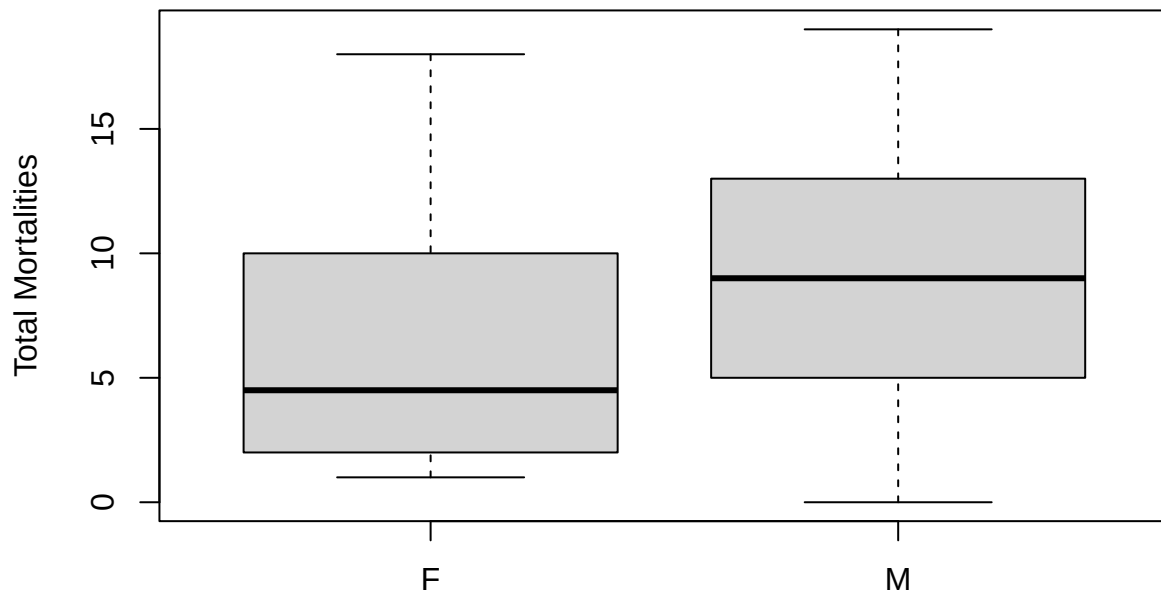
In contrast, the vast majority of male mortality is due to vehicular collisions.

Out of curiosity, we can look for differences in mortality rates for males and females.

```
#Total
t.test(total.female, total.male, paired="T")

##
## Paired t-test
##
## data: total.female and total.male
## t = -3.9403, df = 21, p-value = 0.0007492
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
## -4.444433 -1.373748
## sample estimates:
## mean difference
## -2.909091

boxplot(total.female, total.male,
  names = c("F", "M"),
  ylab = "Total Mortalities")
```

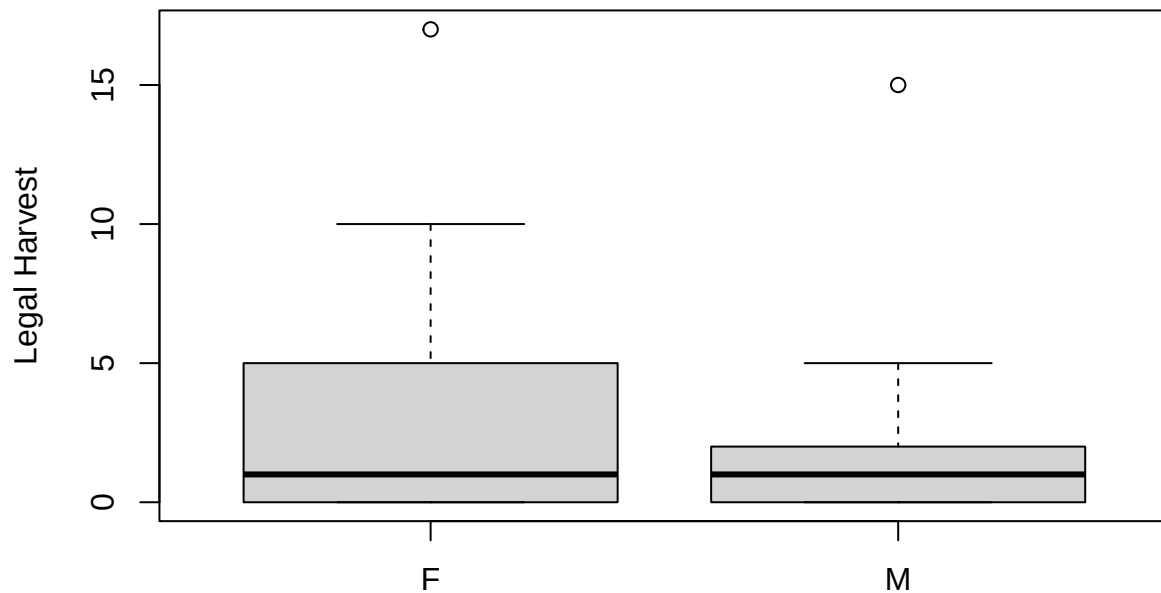


```
#Legal Harvest
```

```
t.test(mort$harvestF, mort$harvestM, paired = "T")
```

```
##
## Paired t-test
##
## data: mort$harvestF and mort$harvestM
## t = 2.6248, df = 21, p-value = 0.01583
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
##  0.2360416 2.0366856
## sample estimates:
## mean difference
##      1.136364
```

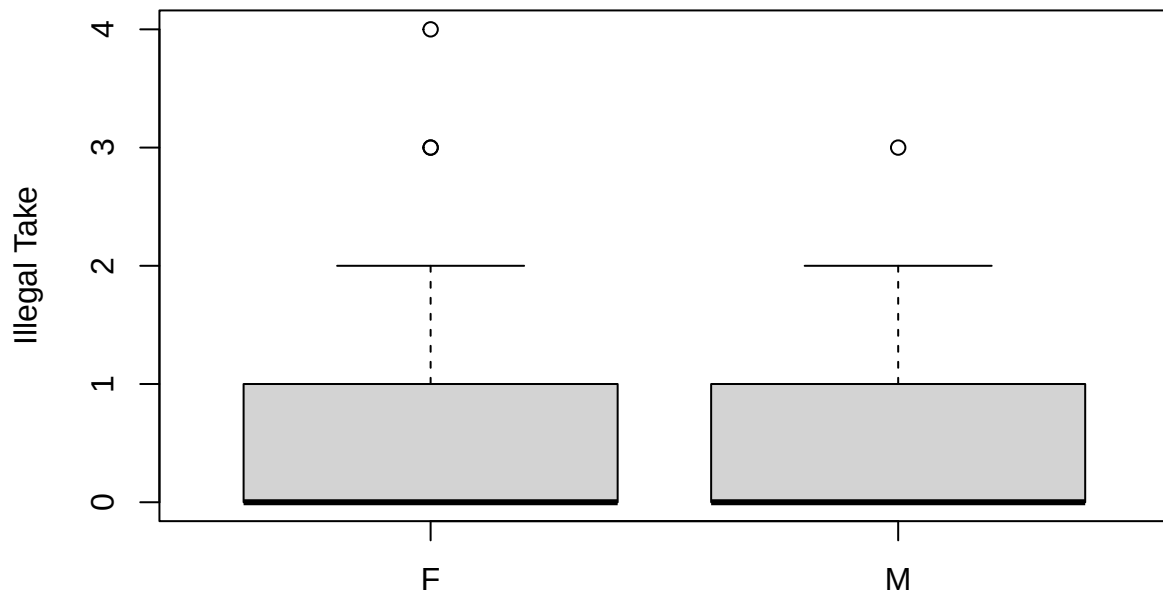
```
boxplot(mort$harvestF, mort$harvestM,
        names = c("F", "M"),
        ylab = "Legal Harvest")
```



```
#Illegal Take
t.test(mort$illegalF, mort$illegalM, paired = "T")
```

```
##
## Paired t-test
##
## data: mort$illegalF and mort$illegalM
## t = 0.64219, df = 21, p-value = 0.5277
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
## -0.5087048 0.9632503
## sample estimates:
## mean difference
## 0.2272727
```

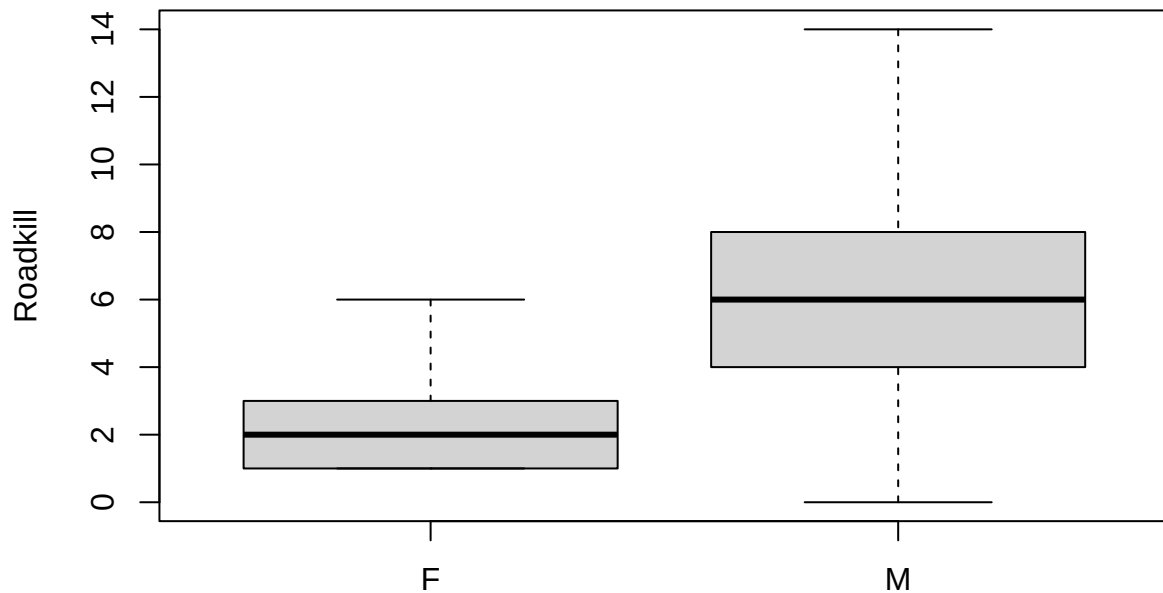
```
boxplot(mort$illegalF, mort$illegalM,
        names=c("F", "M"),
        ylab = "Illegal Take")
```



```
#Roadkill
t.test(mort$roadF, mort$roadM, paired = "T")
```

```
##  
## Paired t-test  
##  
## data: mort$roadF and mort$roadM  
## t = -6.8008, df = 21, p-value = 1.003e-06  
## alternative hypothesis: true mean difference is not equal to 0  
## 95 percent confidence interval:  
## -5.579292 -2.966163  
## sample estimates:  
## mean difference  
## -4.272727
```

```
boxplot(mort$roadF, mort$roadM,
        names = c("F", "M"),
        ylab = "Roadkill")
```

From the quick comparisons above, it appears that total mortality is indeed higher in males. However, legal harvest numbers are higher in females than in males. Perhaps this is due to male bears leaving the study area during the winter. The difference in harvest is more than offset by higher male mortality rates from vehicular collisions.

Average mortality numbers

To compare our abundance estimates to the predictions from simulations in Hooker et al. (2020), we first need to characterize the average total mortality for a few different periods. This way we can identify the scenario that most closely matches the data since the period covered by the Hooker et al. (2020) analysis.

Pre-2012

These numbers won't be directly used, they're included only for the sake of completeness.

```
#Combined
total.all[which(mort$year < 2012)]
```

```
## [1]  2  3 11  6  7 10 13  9 39
```

```
mean(total.all[which(mort$year < 2012)])
```

```
## [1] 11.11111
```

```
median(total.all[which(mort$year < 2012)])
```

```
## [1] 9
```

```
#Females
```

```
total.female[which(mort$year < 2012)]
```

```
## 1 2 3 4 5 6 7 8 9
```

```
## 2 2 3 1 1 1 4 5 18
```

```
mean(total.female[which(mort$year < 2012)])
```

```
## [1] 4.111111
```

```
median(total.female[which(mort$year < 2012)])
```

```
## [1] 2
```

```
#Males
```

```
total.male[which(mort$year < 2012)]
```

```
## 1 2 3 4 5 6 7 8 9
```

```
## 0 1 8 5 5 5 8 3 19
```

```
mean(total.male[which(mort$year < 2012)])
```

```
## [1] 6
```

```
median(total.male[which(mort$year < 2012)])
```

```
## [1] 5
```

2012-2016

Now mortalities for the Hooker et al. (2020) study period.

```
#Combined
```

```
total.all[which(mort$year %in% c(2012:2016))]
```

```
## [1] 22 6 12 25 24
```

```
mean(total.all[which(mort$year %in% c(2012:2016))])
```

```
## [1] 17.8
```

```
median(total.all[which(mort$year %in% c(2012:2016))])
```

```
## [1] 22
```

```
#Females
```

```
total.female[which(mort$year %in% c(2012:2016))]
```

```
## 10 11 12 13 14
```

```
## 11 2 4 12 9
```

```
mean(total.female[which(mort$year %in% c(2012:2016))])
```

```
## [1] 7.6
```

```
median(total.female[which(mort$year %in% c(2012:2016))])
```

```
## [1] 9
```

```
#Males
```

```
total.male[which(mort$year %in% c(2012:2016))]
```

```
## 10 11 12 13 14
```

```
## 11 4 8 12 14
```

```
mean(total.male[which(mort$year %in% c(2012:2016))])
```

```
## [1] 9.8
```

```
median(total.male[which(mort$year %in% c(2012:2016))])
```

```
## [1] 11
```

2017-2024

And for the period since the Hooker study...

```
#Combined
```

```
total.all[which(mort$year %in% c(2017:2024))]
```

```
## [1] 17 23 14 18 26 21 25 20
```

```
mean(total.all[which(mort$year %in% c(2017:2024))])
```

```
## [1] 20.5
```

```
median(total.all[which(mort$year %in% c(2017:2024))])
```

```
## [1] 20.5
```

```
#Females
```

```
total.female[which(mort$year %in% c(2017:2024))]
```

```
## 15 16 17 18 19 20 21 22
```

```
## 3 5 4 7 10 11 10 9
```

```
mean(total.female[which(mort$year %in% c(2017:2024))])
```

```
## [1] 7.375
```

```
median(total.female[which(mort$year %in% c(2017:2024))])
```

```
## [1] 8
```

```
#Males
```

```
total.male[which(mort$year %in% c(2017:2024))]
```

```
## 15 16 17 18 19 20 21 22
```

```
## 14 16 8 10 13 10 14 10
```

```
mean(total.male[which(mort$year %in% c(2017:2024))])
```

```
## [1] 11.875
```

```
median(total.male[which(mort$year %in% c(2017:2024))])
```

```
## [1] 11.5
```

How many additional mortalities?

Comparing recent female mortality numbers to the Hooker et al. (2020) study period...

```
#Hooker period
```

```
mean(total.female[which(mort$year %in% c(2012:2016))])
```

```
## [1] 7.6
```

```
median(total.female[which(mort$year %in% c(2012:2016))])
```

```
## [1] 9
```

```
#post-Hooker period
mean(total.female[which(mort$year %in% c(2017:2024))])
```

```
## [1] 7.375
```

```
median(total.female[which(mort$year %in% c(2017:2024))])
```

```
## [1] 8
```

```
#In comparisons below
# positive numbers mean lower contemporary mortality!

#Difference b/w '12-'16 mean & recent mean
mean(total.female[which(mort$year %in% c(2012:2016))]) -
  mean(total.female[which(mort$year %in% c(2017:2024))])
```

```
## [1] 0.225
```

```
median(total.female[which(mort$year %in% c(2012:2016))]) -
  median(total.female[which(mort$year %in% c(2017:2024))])
```

```
## [1] 1
```

```
#Difference b/w '12-'16 mean & max recent annual mortality?
mean(total.female[which(mort$year %in% c(2012:2016))]) -
  max(total.female[which(mort$year %in% c(2017:2024))])
```

```
## [1] -3.4
```

```
median(total.female[which(mort$year %in% c(2012:2016))]) -
  max(total.female[which(mort$year %in% c(2017:2024))])
```

```
## [1] -2
```

```
#Annual difference to '12-'16 means
mean(total.female[which(mort$year %in% c(2012:2016))]) -
  total.female[which(mort$year %in% c(2017:2024))]
```

```
## 15 16 17 18 19 20 21 22
## 4.6 2.6 3.6 0.6 -2.4 -3.4 -2.4 -1.4
```

```
median(total.female[which(mort$year %in% c(2012:2016))]) -
  total.female[which(mort$year %in% c(2017:2024))]
```

```
## 15 16 17 18 19 20 21 22
## 6 4 5 2 -1 -2 -1 0
```

```
#Individual sources
#Legal harvest?
mean(mort$harvestF[which(mort$year %in% c(2012:2016))]) -
  mean(mort$harvestF[which(mort$year %in% c(2017:2024))])
```

```
## [1] 2.125
```

```
median(mort$harvestF[which(mort$year %in% c(2012:2016))]) -
  median(mort$harvestF[which(mort$year %in% c(2017:2024))])
```

```
## [1] 3
```

```
#Roadkill?
mean(mort$roadF[which(mort$year %in% c(2012:2016))]) -
  mean(mort$roadF[which(mort$year %in% c(2017:2024))])
```

```
## [1] -1
```

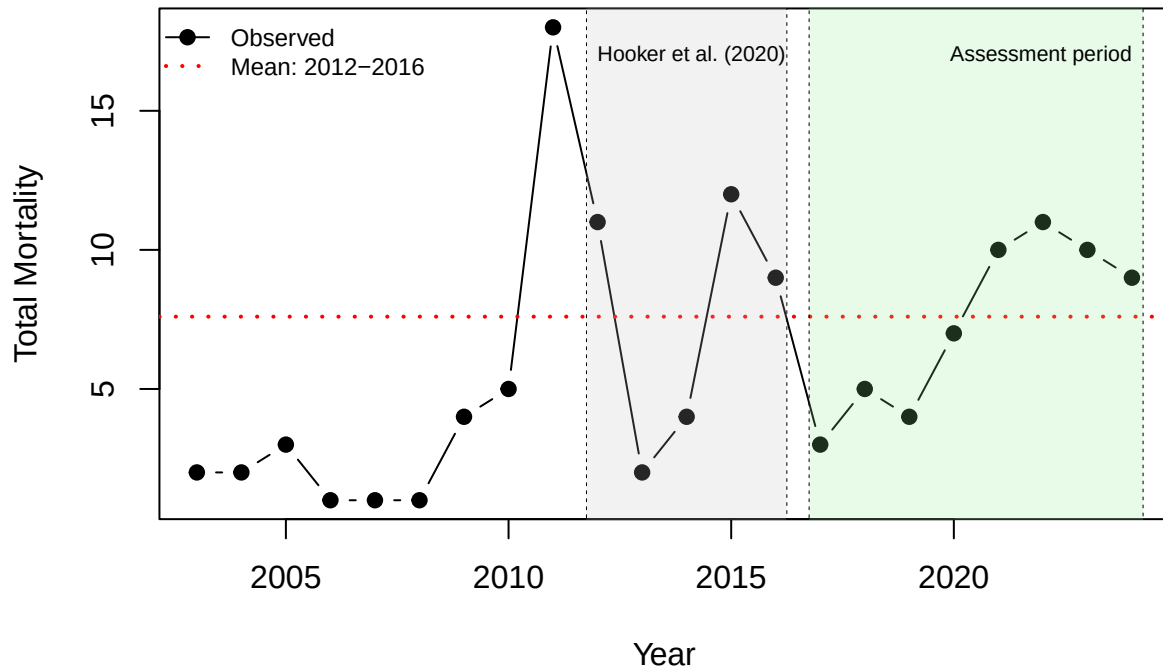
```
median(mort$roadF[which(mort$year %in% c(2012:2016))]) -
  median(mort$roadF[which(mort$year %in% c(2017:2024))])
```

```
## [1] -1
```

In fact, mean and median female mortality numbers since 2017 are lower than they were during the Hooker et al. (2020) study period. The *maximum* single year total mortality of females since the Hooker et al. (2020) study is **3.4 bears** above the mean for their study period. Notably, annual female mortality since 2017 has exceeded the average from the Hooker et al. (2020) study period in 4 years (the last years considered, 2021-2024).

```
plot(total.female[which(mort$year < 2025)] ~ mort$year[which(mort$year < 2025)],
     type = "b", lwd = 1, col="black", pch=19,
     xlab = "Year", ylab = "Total Mortality", main = "Females")
abline(h = mean(total.female[which(mort$year %in% c(2012:2016))]),
       lty = 3, col="red", lwd = 2)
legend("topleft", bty="n", cex = 0.75,
       lty=c(1,3), pch=c(19,19), pt.cex=c(1,0), lwd = c(1,2),
       col = c("black","red"),
       legend = c("Observed","Mean: 2012-2016"))
polygon(x=c(2011.75,2011.75,2016.25,2016.25),y=c(-100,100,100,-100),
       col=alpha("grey",0.2), lty=2, lwd = 0.5)
text(x=2012, y=17, labels="Hooker et al. (2020)", cex=0.65, adj=0)
polygon(x=c(2016.75,2016.75,2024.25,2024.25),y=c(-100,100,100,-100),
       col=alpha("lightgreen",0.2), lty=2, lwd = 0.5)
text(x=2024, y=17, labels="Assessment period", cex=0.65, adj=1)
```

Females



```
#Number of additional mortalities, above 2012–2016 average, for 2021–2024
total.female[which(mort$year %in% c(2021:2024))] -
  mean(total.female[which(mort$year %in% c(2012:2016))])
```

```
## 19 20 21 22
## 2.4 3.4 2.4 1.4
```

In terms of the two major sources of mortality, average legal harvests since 2017 are down by 2.125 females, while average vehicular collisions are up by 1 females.

Assess PVA predictions

Given the female mortality observed since the Hooker et al. (2020) study period, and the lower mean annual harvest numbers, management of the central GA black bear harvest has been effective at limiting annual female mortality.

The average total *known* female mortality since the Hooker et al. (2020) study (7.375) was below the average total mortality from the Hooker et al. (2020) study period (7.6), while the maximum single-year mortality since 2016 (11 females in 2022) was slightly higher. Thus, observed total *known* female mortality in central GA is most consistent with the two lowest harvest rate scenarios from Hooker et al. (2020), *status quo* or $h = 5$ expected additional females harvested per year.

Under both of these scenarios, 50-year extinction risks in the PVA remained below 1%, and population size was relatively stable (Hooker et al. 2020). Quoting from their Discussion section:

From an abundance of 165 (95% CI = 134–206) in 2016, the 50-year population forecasts (i.e., estimated abundance in the year 2067) for each of the 5 harvest scenarios in order of lowest to highest harvest were 169 (95% CI = 134–268), 159 (95% CI = 124–243), 147 (95% CI = 103–223), 132 (95% CI = 0–197), and 105 (95% CI = 0–174). Extinction probabilities for the status quo scenario (i.e., the lowest level of harvest of any of the scenarios) was negligible at <0.001% with no abundance projections falling to zero during the 50-year timeline. Similarly, the scenario in which the status quo harvest was increased by 5 females had few projections in which abundance dropped to zero, resulting in an extinction risk of <1%. –Hooker et al. (2020)

Re-fit the Hooker model

To quantify the degree to which our estimates of female abundance align with predictions from the population viability analyses of Hooker et al. (2020), we will first re-fit the model to capture data for female bears. The files needed to do this are available on the Hooker et al. (2020) [GitHub repository](#): `dT-s0-rPDD2-phi0-pYB-sig0.jag`, `fit.R`, `ga_bear_data_females.gzip`, and `state-space360.tif`.

```
#Code below is copied from fit.R
# paths to the JAGS model, capture history data, and state space file
# are modified slightly
```

```
## Fit open population SCR models to central GA black bear data
## Female-only analysis
```

```
library(rjags)
```

```
## Loading required package: coda
```

```
## Linked to JAGS 4.3.2
```

```
## Loaded modules: basemod,bugs
```

```
library(lattice)
library(raster)
```

```
## Loading required package: sp
```

```
list.files()
```

```
## [1] "14_assessPVA_files" "14_assessPVA.Rmd"  "assessPVA.o"
## [4] "assessPVA.sh"      "render_assessPVA.R"
```

```
## Load SCR data
load("../input/fromRepo/ga_bear_data_females.gzip")
```

```
## Inspect the loaded data objects
str(ch4d.f)
```



```
## num [1:141, 1:173, 1:8, 1:5] 0 0 0 0 0 0 0 0 0 0 ...
## - attr(*, "dimnames")=List of 4
## ..$ : chr [1:141] "GA1-01-031-01T" "GA1-01-056-01T" "GA1-01-066-02B" "GA1-01-075-02B" ...
## ..$ : chr [1:173] "x258787-y3600079" "x260710-y3599587" "x265200-y3601052" "x266238-y3600529" ...
## ..$ : NULL
## ..$ : chr [1:5] "2012" "2013" "2014" "2015" ...
```

```
str(traps)
```

```
## int [1:173, 1:2] 258787 260710 265200 266238 267777 269347 271377 272584 274249 265076 ...
## - attr(*, "dimnames")=List of 2
## ..$ : chr [1:173] "x258787-y3600079" "x260710-y3599587" "x265200-y3601052" "x266238-y3600529" ...
## ..$ : chr [1:2] "x" "y"
```

```
## Check that traps are in the same order in both objects
## Must be TRUE
all(rownames(traps)==dimnames(ch4d.f)[[3]])
```

```
## [1] TRUE
```

```
all(rownames(oper)==rownames(traps))
```

```
## [1] TRUE
```

```
## Format data
```

```
y <- ch4d.f ## 4D encounter histories
str(y)
```

```
## num [1:141, 1:173, 1:8, 1:5] 0 0 0 0 0 0 0 0 0 0 ...
## - attr(*, "dimnames")=List of 4
## ..$ : chr [1:141] "GA1-01-031-01T" "GA1-01-056-01T" "GA1-01-066-02B" "GA1-01-075-02B" ...
## ..$ : chr [1:173] "x258787-y3600079" "x260710-y3599587" "x265200-y3601052" "x266238-y3600529" ...
## ..$ : NULL
## ..$ : chr [1:5] "2012" "2013" "2014" "2015" ...
```

```
y.dim <- dim(y)
nBears <- y.dim[1]
nTraps <- y.dim[2]
nWeeks <- y.dim[3]
nYears <- y.dim[4]
M <- 500 # Data aug. Maximum number of bears that could have been alive

all(rownames(prevcap.f)==rownames(y)) ## Must be TRUE
```

```
## [1] TRUE
```

```
## Non-spatial, robust design format
y.nonsp.robust <- apply(y>0, c(1,3,4), any)*1L
str(y.nonsp.robust)
```

```

## int [1:141, 1:8, 1:5] 1 1 1 1 1 1 1 0 0 0 ...
## - attr(*, "dimnames")=List of 3
## ..$ : chr [1:141] "GA1-01-031-01T" "GA1-01-056-01T" "GA1-01-066-02B" "GA1-01-075-02B" ...
## ..$ : NULL
## ..$ : chr [1:5] "2012" "2013" "2014" "2015" ...

## Non-spatial, non-robust format
y.nonsp.nonrobust <- apply(y.nonsp.robust>0,
                           c(1,3), any)*1L
str(y.nonsp.nonrobust)

## int [1:141, 1:5] 1 1 1 1 1 1 1 1 1 ...
## - attr(*, "dimnames")=List of 2
## ..$ : chr [1:141] "GA1-01-031-01T" "GA1-01-056-01T" "GA1-01-066-02B" "GA1-01-075-02B" ...
## ..$ : chr [1:5] "2012" "2013" "2014" "2015" ...

## Spatial, non-robust format
y.sp.nonrobust <- apply(y>0, c(1,2,4), any)*1L
str(y.sp.nonrobust)

## int [1:141, 1:173, 1:5] 0 0 0 0 0 0 0 0 0 0 ...
## - attr(*, "dimnames")=List of 3
## ..$ : chr [1:141] "GA1-01-031-01T" "GA1-01-056-01T" "GA1-01-066-02B" "GA1-01-075-02B" ...
## ..$ : chr [1:173] "x258787-y3600079" "x260710-y3599587" "x265200-y3601052" "x266238-y3600529" ...
## ..$ : chr [1:5] "2012" "2013" "2014" "2015" ...

## First and last detections
first.det <- apply(y.nonsp.nonrobust>0, 1, function(x) min(which(x)))
last.det <- apply(y.nonsp.nonrobust>0, 1, function(x) max(which(x)))
known.alive <- last.det-first.det+1
table(known.alive)

## known.alive
## 1 2 3 4 5
## 80 17 13 17 14

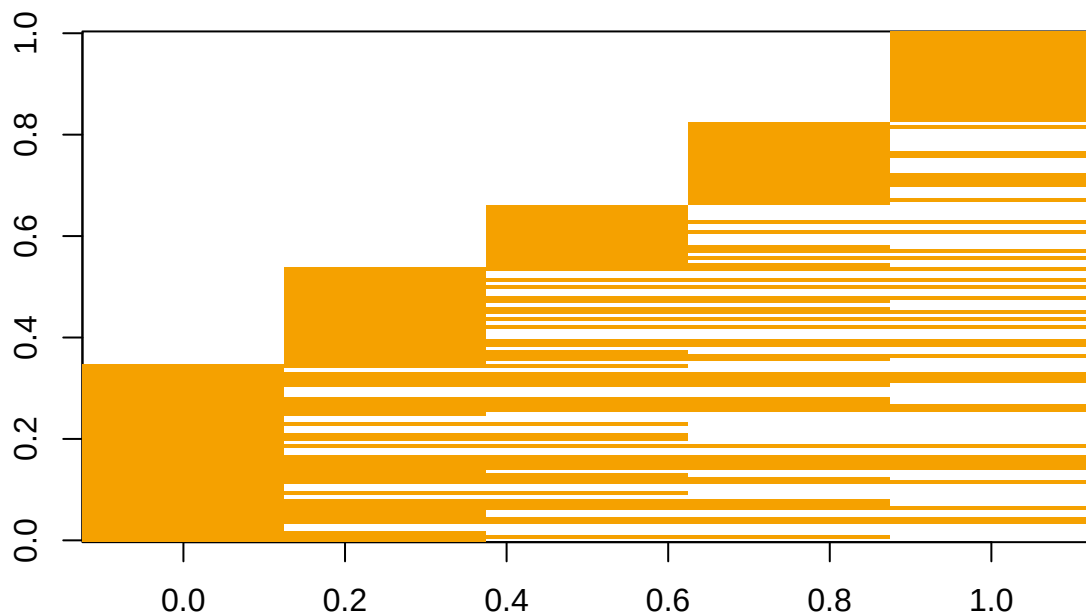
## Data augmentation
yaug <- array(0, c(M, nTraps, nWeeks, nYears))
yaug[1:nBears,,] <- y
str(yaug)

## num [1:500, 1:173, 1:8, 1:5] 0 0 0 0 0 0 0 0 0 0 ...

## Known values of the alive/dead state matrix
zdata.aug <- matrix(NA, M, nYears)
for(i in 1:nBears) {
  zdata.aug[i,first.det[i]:last.det[i]] <- 1
}

## Visualize the observed information about lifetime
image(t(zdata.aug[1:nBears,]))

```



```
## Bears known to be alive in each year
colSums(zdata.aug[1:nBears,], na.rm=TRUE)
```

```
## [1] 49 60 58 63 61
```

```
## Operational status of traps
nTrapsYr <- colSums(oper)
nTrapsYr
```

```
## 2012 2013 2014 2015 2016
## 126 129 135 135 134
```

```
oper.traps <- matrix(0L, nTraps, nYears)
for(i in 1:nYears) {
  oper.traps[1:nTrapsYr[i],i] <- which(oper[,i]==1)
}
```

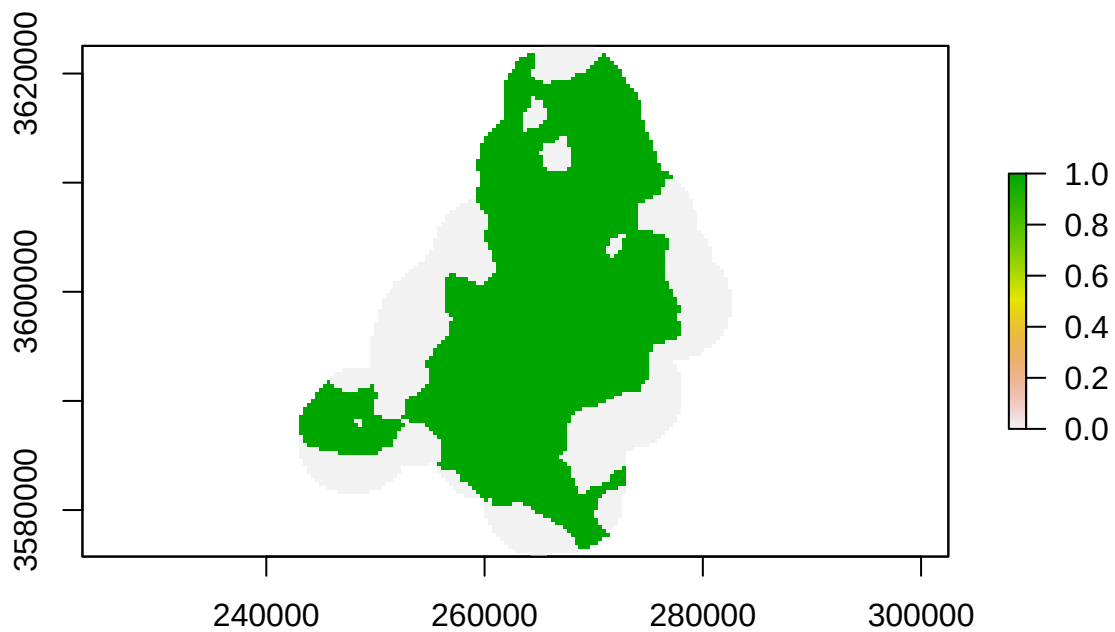
```
## State-space
x.range <- range(traps[,1])
y.range <- range(traps[,2])
buffer <- 4000
xlim0 <- x.range+c(-buffer, buffer)
```

```
ylim0 <- y.range+c(-buffer, buffer)

S360 <- raster("../input/fromRepo/state-space360.tif")

traps.in <- extract(S360, traps)

plot(S360)
```



```
## Convert raster to a data.frame
S360dat <- as.data.frame(S360, xy=TRUE)
names(S360dat) <- c("x", "y", "S360")
S360dat$x <- floor(S360dat$x)
S360dat$y <- floor(S360dat$y)
table(diff(S360dat$x))
```

```
##
## -39240    360
##    129  14170
```

```
table(diff(S360dat$y))
```

```
##
## -360      0
##    129 14170
```

```

nPix360 <- nrow(S360dat)
dist360 <- matrix(NA, nPix360, nTraps)
for(j in 1:nTraps) {
  dist360[,j] <- sqrt((S360dat$x-traps[j,1])^2 +
                     (S360dat$y-traps[j,2])^2)
}

S360dat$pixIn <- (rowSums(dist360<buffer)>0)*1L
table(S360dat$pixIn)

```

```

##
##      0      1
## 7621 6679

```

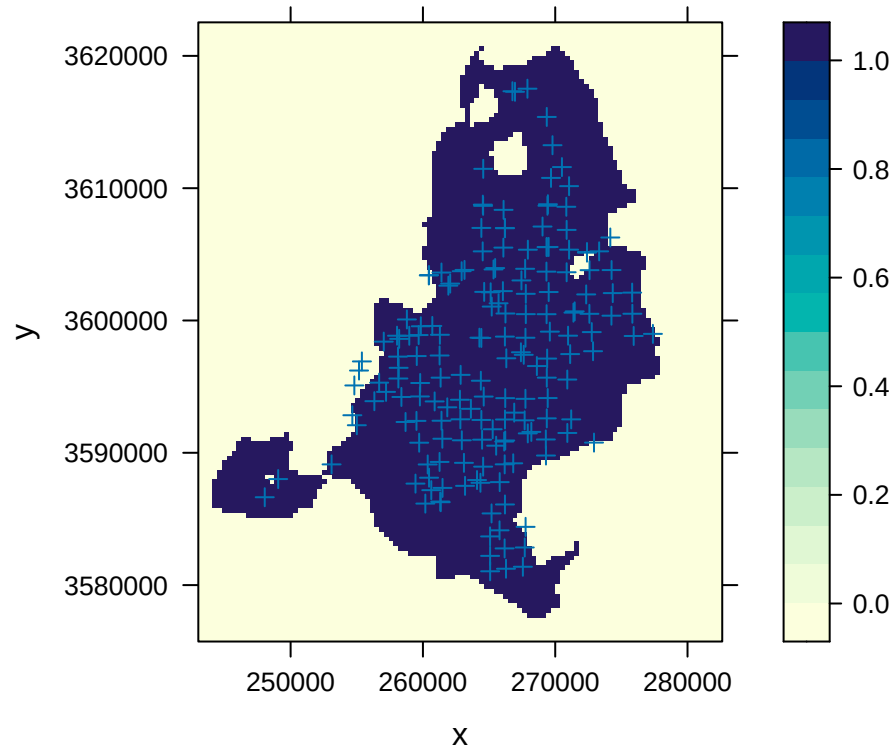
```

dim.S360 <- dim(S360)

S360dat$habitat <- ifelse((S360dat$S360>0) &
                        (S360dat$pixIn>0),
                        1L, 0L)

## Visualize traps and state space
levelplot(habitat ~ x+y, S360dat, aspect='iso',
  panel=function(...) {
    panel.levelplot(...)
    lpoints(traps, pch=3)
  })

```



```

delta <- res(S360)[1]
xseq <- sort(unique(S360dat$x))
yseq <- sort(unique(S360dat$y), decreasing=TRUE)
xlim <- range(xseq)+c(-delta/2,delta/2)
ylim <- range(yseq)+c(-delta/2,delta/2)

dim.S360

## [1] 130 110 1

range(ceiling((runif(1e5, xlim[1], xlim[2])-xlim[1])/delta))

## [1] 1 110

range(ceiling((ylim[2]-runif(1e5, ylim[1], ylim[2]))/delta))

## [1] 1 130

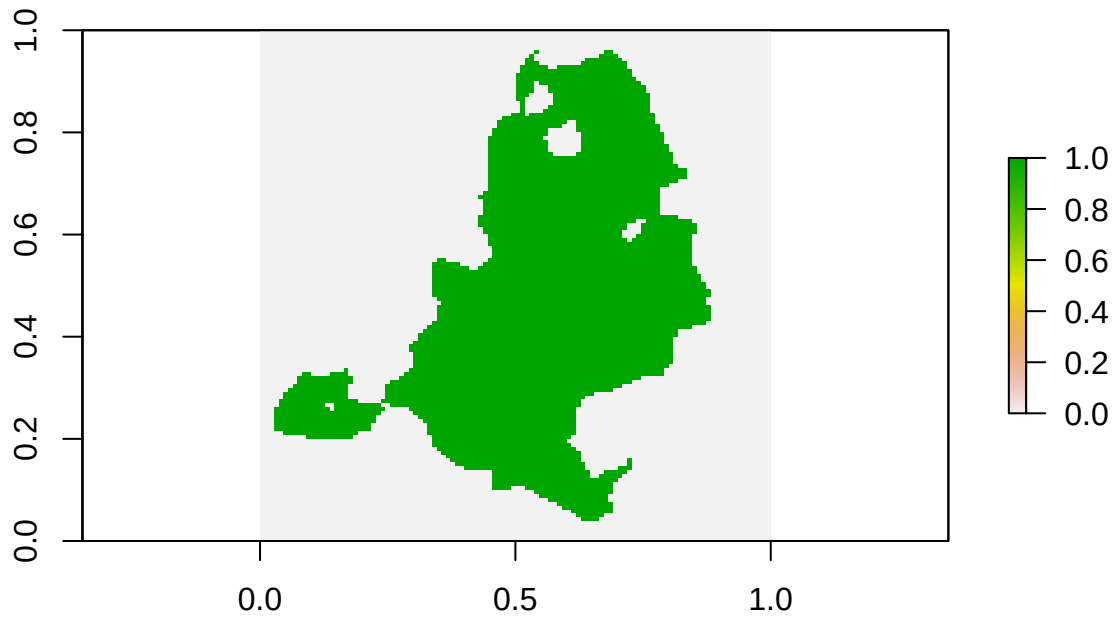
S <- matrix(S360dat$habitat, dim.S360[1], dim.S360[2],
            dimnames=list(yseq, xseq), byrow=TRUE)
str(S)

## int [1:130, 1:110] 0 0 0 0 0 0 0 0 0 0 ...

```

```
## - attr(*, "dimnames")=List of 2
## ..$ : chr [1:130] "3622355" "3621995" "3621635" "3621275" ...
## ..$ : chr [1:110] "243204" "243564" "243924" "244284" ...
```

```
plot(raster(S))
```



```
sum(S)*delta^2 / 1e6 ## Area in km-sq
```

```
## [1] 622.5984
```

```
## JAGS
## Models are slow. About 450 iter/hr
```

```
library(rjags)
```

```
### Put the data in a list
```

```
jd <- list(y=y, y.zero=matrix(0, M, nYears),
           z=zdata.aug,
           nBears=nBears, M=M,
           nTrapsYr=as.integer(nTrapsYr),
           nWeeks=nWeeks, nYears=nYears, nFuture=0,
           oper.traps=oper.traps,
```

```

        prevcap=prevcap.f+1,
        S=S, delta=delta,
        ones=matrix(1, M, nYears),
        x=traps, xlim=xlim, ylim=ylim)
jd$z=cbind(zdata.aug, matrix(NA, M, jd$nFuture))

str(jd)

```

```

## List of 17
## $ y      : num [1:141, 1:173, 1:8, 1:5] 0 0 0 0 0 0 0 0 0 0 ...
##   ..- attr(*, "dimnames")=List of 4
##   .. ..$ : chr [1:141] "GA1-01-031-01T" "GA1-01-056-01T" "GA1-01-066-02B" "GA1-01-075-02B" ...
##   .. ..$ : chr [1:173] "x258787-y3600079" "x260710-y3599587" "x265200-y3601052" "x266238-y3600529" .
##   .. ..$ : NULL
##   .. ..$ : chr [1:5] "2012" "2013" "2014" "2015" ...
## $ y.zero   : num [1:500, 1:5] 0 0 0 0 0 0 0 0 0 0 ...
## $ z        : num [1:500, 1:5] 1 1 1 1 1 1 1 1 1 1 ...
## $ nBears   : int 141
## $ M        : num 500
## $ nTrapsYr : int [1:5] 126 129 135 135 134
## $ nWeeks   : int 8
## $ nYears   : int 5
## $ nFuture  : num 0
## $ oper.traps: int [1:173, 1:5] 1 2 3 4 5 6 7 8 9 10 ...
## $ prevcap  : num [1:141, 1:173, 1:8, 1:5] 1 1 1 1 1 1 1 1 1 1 ...
##   ..- attr(*, "dimnames")=List of 4
##   .. ..$ : chr [1:141] "GA1-01-031-01T" "GA1-01-056-01T" "GA1-01-066-02B" "GA1-01-075-02B" ...
##   .. ..$ : chr [1:173] "x258787-y3600079" "x260710-y3599587" "x265200-y3601052" "x266238-y3600529" .
##   .. ..$ : NULL
##   .. ..$ : chr [1:5] "2012" "2013" "2014" "2015" ...
## $ S        : int [1:130, 1:110] 0 0 0 0 0 0 0 0 0 0 ...
##   ..- attr(*, "dimnames")=List of 2
##   .. ..$ : chr [1:130] "3622355" "3621995" "3621635" "3621275" ...
##   .. ..$ : chr [1:110] "243204" "243564" "243924" "244284" ...
## $ delta    : num 360
## $ ones     : num [1:500, 1:5] 1 1 1 1 1 1 1 1 1 1 ...
## $ x        : int [1:173, 1:2] 258787 260710 265200 266238 267777 269347 271377 272584 274249 265070
##   ..- attr(*, "dimnames")=List of 2
##   .. ..$ : chr [1:173] "x258787-y3600079" "x260710-y3599587" "x265200-y3601052" "x266238-y3600529" .
##   .. ..$ : chr [1:2] "x" "y"
## $ xlim     : num [1:2] 243024 282624
## $ ylim     : num [1:2] 3575735 3622535

```

Function to generate initial values

```

ji <- function() {
  nFuture <- jd$nFuture
  nBears <- jd$nBears
  M <- jd$M
  nYears <- jd$nYears
  nWeeks <- jd$nWeeks
  nFuture <- jd$nFuture
  traps <- jd$x
  nTraps <- nrow(traps)

```



```

zi <- matrix(0, M, nYears+nFuture)
## Avoid N[t]=0, which causes problems with initial values of lambda
if(nFuture>0)
  zi[(M-5):M,(nYears+1):(nYears+nFuture)] <- 1
si1 <- traps[sample.int(nTraps, size=M, replace=TRUE,
                        prob=traps.in),]
si.rec <- si <- array(NA, c(M, 2, nYears))
si[, ,1] <- si1
si.rec[, ,1] <- si1
for(i in 1:nBears) {
  zi[i,first.det[i]:last.det[i]] <- NA
  traplocs.i <- (rowSums(jd$y[i,,])>0) * traps.in
  if(all(!traplocs.i))
    next
  si[i,,1] <- colMeans(traps[traplocs.i,,drop=FALSE])
  si.rec[i,,1] <- colMeans(traps[traplocs.i,,drop=FALSE])
}
p0i <- array(NA, c(nWeeks, nYears, 2))
p0i[, ,2] <- matrix(runif(nWeeks*nYears), nWeeks, nYears)
out <- list(z=zi, s=si, s.rec=si.rec, p0=p0i,
            sigma=runif(1, 3000, 4000), behave=runif(1, 0.5, 1),
            mu.lphi=0, sig.lphi=.01,
            psi=c(runif(1), rep(NA, nYears+nFuture-1)),
            igamma0=runif(1, 0.2, 0.3), gamma1=runif(1, 0, 0.001))
out$".RNG.name" <- "base::Mersenne-Twister"
out$".RNG.seed" <- sample.int(1e6, 1)
return(out)
}

str(ji())

```

```

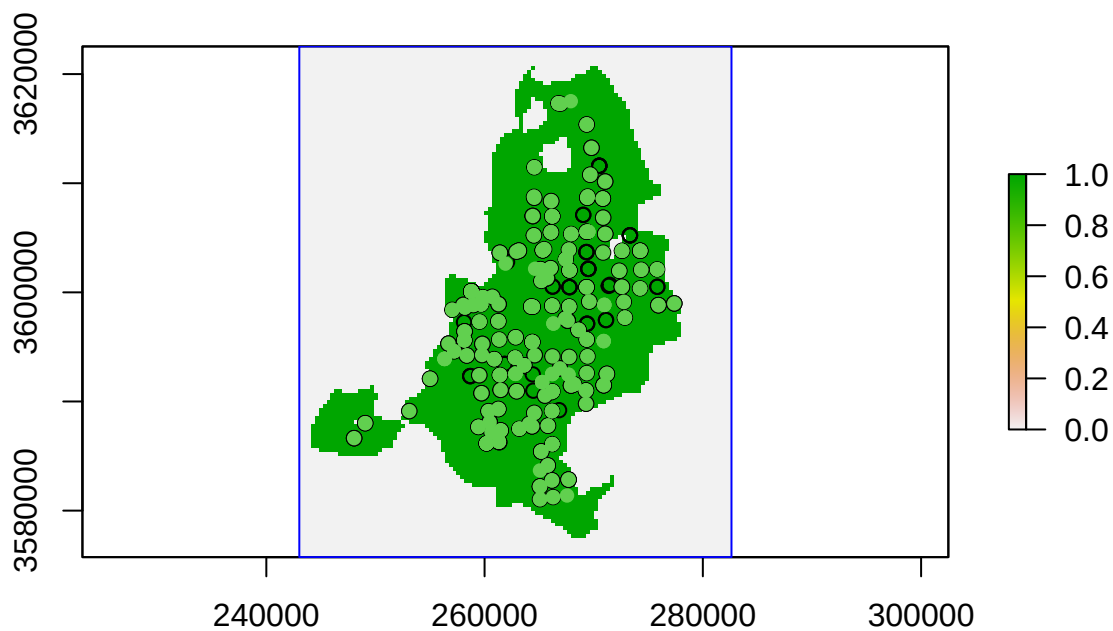
## List of 13
## $ z      : num [1:500, 1:5] NA NA NA NA NA NA NA NA NA NA ...
## $ s      : num [1:500, 1:2, 1:5] 258787 258787 258787 258787 258787 ...
## $ s.rec   : num [1:500, 1:2, 1:5] NA NA NA NA NA NA NA NA NA NA ...
## $ p0      : num [1:8, 1:5, 1:2] NA NA NA NA NA NA NA NA NA NA ...
## $ sigma   : num 3278
## $ behave  : num 0.828
## $ mu.lphi : num 0
## $ sig.lphi : num 0.01
## $ psi     : num [1:5] 0.184 NA NA NA NA
## $ igamma0 : num 0.243
## $ gamma1  : num 0.000758
## $ .RNG.name: chr "base::Mersenne-Twister"
## $ .RNG.seed: int 240748

```

```

## Visualize initial values of activity centers
plot(raster(S, xmn=xlim[1], ymn=ylim[1], xmx=xlim[2], ymx=ylim[2]))
points(ji()$s[, ,1])
points(ji()$s.rec[, ,2], pch=16, col=3)
rect(xlim[1], ylim[1], xlim[2], ylim[2], border="blue")

```



```
#symbols(Sx, Sy, circles=rep(buffer, length(Sx)),
#        fg="blue", add=TRUE, inches=FALSE)

#### Parameters to monitor
jp <- c("p0", "sigma", "behave",
        "igamma0", "gamma0", "gamma1", "phi", "gamma", "EN",
        "N", "Ntotal", "survivors", "recruits", "lambda")
```

```
## Models in parallel with WAIC computation
```

```
library(parallel)
```

```
(nCores <- max(1, min(detectCores()-1, 4)))
```

```
## [1] 4
```

```
cl1 <- makeCluster(nCores)
```

```
clusterExport(cl1, c("jd", "ji", "jp",
                     "traps.in", "first.det", "last.det"))
```

```
## Compile and adapt. Comment out next line if you don't want reproducible results
clusterSetRNGStream(cl1, iseed=389761)
```

```

system.time({
  jm.out <- clusterEvalQ(c11, {
    library(rjags)
    load.module("dic")
    load.module("glm")
    jm <- jags.model(file="../input/fromRepo/dT-s0-rPDD2-phi0-pYB-sig0.jag",
                     data=jd, inits=ji,
                     n.chains=1, n.adapt=500)

    return(jm)
  })
}) ## Approx 1.5 hr

```

```

##      user      system elapsed
##    0.537      0.072 3901.696

```

```

save(jm.out, file="jm.gzip")

## Draw posterior samples, but don't monitor latent variables
system.time({
  jc1.out <- clusterEvalQ(c11, {
    jc1 <- coda.samples(model=jm,
                       variable.names=c(jp, "deviance"),
                       n.iter=5000)

    return(as.mcmc(jc1))
  })
}) ## 450 iterations/hr

```

```

##      user      system elapsed
##    0.003      0.133 28768.900

```

```

save(jc1.out, file="jc1.gzip")

## mc1 <- as.mcmc.list(jc1.out)
## plot(mc1, ask=TRUE)

## Draw some more samples, including the latent variables needed
## for forecasting
jc2.out <- clusterEvalQ(c11, {
  jc2 <- coda.samples(model=jm,
                     variable.names=c(jp, "deviance", "z", "a", "s"),
                     n.iter=5000, n.thin=1)

  return(as.mcmc(jc2))
})

save(jc2.out, file="jc2.gzip")

rm(jc1.out, jc2.out)
gc()

```

```
##          used (Mb) gc trigger (Mb) max used (Mb)
## Ncells  2851494 152.3   4528244  241.9   4528244  241.9
## Vcells 28716779 219.1  225630090 1721.5 233571896 1782.1
```

```
## WAIC
js3.out <- clusterEvalQ(cl1, {
  js3 <- jags.samples(model=jm,
    variable.names=c("deviance","WAIC"),
    n.iter=5000, n.thin=1, type="mean")
  return(js3)
})

save(js3.out, file="js3.gzip")

## These additional samples aren't really necessary because the Markov
## chains usually converge very rapidly
jc4.out <- clusterEvalQ(cl1, {
  jc4 <- coda.samples(model=jm,
    variable.names=c("deviance", "z", "a", "s"),
    n.iter=5000, n.thin=1)
  return(as.mcmc(jc4))
})

save(jc4.out, file="jc4.gzip")
```

The `fit.R` script continues, but the samples from the posterior distribution saved in `jc4.gzip` are what we need for the next steps, so we'll stop here.

Now summarize the posterior distribution, focusing on abundance and survival.

```
library(coda)
summary(as.mcmc.list(jc4.out)[,c("N[1]", "N[2]", "N[3]", "N[4]", "N[5]", "phi")])
```

```
##
## Iterations = 15501:20500
## Thinning interval = 1
## Number of chains = 4
## Sample size per chain = 5000
##
## 1. Empirical mean and standard deviation for each variable,
##    plus standard error of the mean:
##
##      Mean      SD Naive SE Time-series SE
## N[1] 106.346 12.52392 0.0885575      0.3216121
## N[2] 135.132 12.57004 0.0888836      0.3991162
## N[3] 152.501 14.67288 0.1037529      0.5499559
```

```
## N[4] 159.799 16.47322 0.1164833 0.6528244
## N[5] 166.781 18.83846 0.1332080 0.7178218
## phi 0.754 0.03433 0.0002427 0.0008824
##
## 2. Quantiles for each variable:
##
##      2.5%      25%      50%      75%      97.5%
## N[1] 85.0000 97.0000 105.0000 114.0000 134.0000
## N[2] 112.0000 126.0000 135.0000 143.0000 161.0000
## N[3] 126.0000 142.0000 152.0000 162.0000 184.0000
## N[4] 130.0000 148.0000 159.0000 170.0000 195.0000
## N[5] 133.0000 153.0000 166.0000 179.0000 206.0000
## phi 0.6859 0.7309 0.7545 0.7779 0.8195
```

Forecast abundance

We then use the `forecast_fn.R` and `forecast.R` scripts from the [GitHub repository](#) associated with that project to recreate the forecasted population sizes.

```
#Code below is copied from forecast.R
# the path to the forecast_fn.R script is modified slightly
```

```
## Forecast Central GA black bear dynamics and viability
```

```
library(coda)
```

```
source("../input/fromRepo/forecast_fn.R")
```

```
args(forecast)
```

```
## function (mcout, nFuture, M, harvest = NULL, stochastic = FALSE,
##      season = c("postrepro", "prerepro"), maxRecruit = Inf, NbarSims = 1,
##      storeZ = FALSE, report = Inf)
## NULL
```

```
## Load posterior samples
load("jc4.gzip")
jc4 <- as.mcmc.list(jc4.out)
ps <- window(jc4, thin=10)

niter(ps)*nchain(ps)
```

```
## [1] 2000
```

```
## debugonce(forecast)
```

```
## No harvest
system.time({
  forc.h0 <- forecast(ps, nFuture=50, M=15000,
                      harvest=0, stochastic=TRUE,
                      NbarSims=100,
```

})

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```
## iter 1400 of 2000
```



```
## Warning in forecast(ps, nFuture = 50, M = 15000, harvest = 0, stochastic =
## TRUE, : Entrance probability >1. Increase M. For now, N fixed at max.
## Warning in forecast(ps, nFuture = 50, M = 15000, harvest = 0, stochastic =
## TRUE, : Entrance probability >1. Increase M. For now, N fixed at max.
## Warning in forecast(ps, nFuture = 50, M = 15000, harvest = 0, stochastic =
## TRUE, : Entrance probability >1. Increase M. For now, N fixed at max.
## Warning in forecast(ps, nFuture = 50, M = 15000, harvest = 0, stochastic =
## TRUE, : Entrance probability >1. Increase M. For now, N fixed at max.
## Warning in forecast(ps, nFuture = 50, M = 15000, harvest = 0, stochastic =
## TRUE, : Entrance probability >1. Increase M. For now, N fixed at max.
## Warning in forecast(ps, nFuture = 50, M = 15000, harvest = 0, stochastic =
## TRUE, : Entrance probability >1. Increase M. For now, N fixed at max.
## Warning in forecast(ps, nFuture = 50, M = 15000, harvest = 0, stochastic =
## TRUE, : Entrance probability >1. Increase M. For now, N fixed at max.
## Warning in forecast(ps, nFuture = 50, M = 15000, harvest = 0, stochastic =
## TRUE, : Entrance probability >1. Increase M. For now, N fixed at max.
## Warning in forecast(ps, nFuture = 50, M = 15000, harvest = 0, stochastic =
## TRUE, : Entrance probability >1. Increase M. For now, N fixed at max.
## Warning in forecast(ps, nFuture = 50, M = 15000, harvest = 0, stochastic =
## TRUE, : Entrance probability >1. Increase M. For now, N fixed at max.
## Warning in forecast(ps, nFuture = 50, M = 15000, harvest = 0, stochastic =
## TRUE, : Entrance probability >1. Increase M. For now, N fixed at max.
## Warning in forecast(ps, nFuture = 50, M = 15000, harvest = 0, stochastic =
## TRUE, : Entrance probability >1. Increase M. For now, N fixed at max.
## Warning in forecast(ps, nFuture = 50, M = 15000, harvest = 0, stochastic =
## TRUE, : Entrance probability >1. Increase M. For now, N fixed at max.
## Warning in forecast(ps, nFuture = 50, M = 15000, harvest = 0, stochastic =
## TRUE, : Entrance probability >1. Increase M. For now, N fixed at max.
## Warning in forecast(ps, nFuture = 50, M = 15000, harvest = 0, stochastic =
## TRUE, : Entrance probability >1. Increase M. For now, N fixed at max.
## Warning in forecast(ps, nFuture = 50, M = 15000, harvest = 0, stochastic =
## TRUE, : Entrance probability >1. Increase M. For now, N fixed at max.
```

```
## iter 1500 of 2000
## iter 1600 of 2000
## iter 1700 of 2000
## iter 1800 of 2000
## iter 1900 of 2000
## iter 2000 of 2000
```

```
## user system elapsed
## 10365.985 27.248 10439.043
```

```
save(forc.h0, file="forc-h0.gzip")
```

```
## 5 additional females per year
system.time({
  forc.h5 <- forecast(ps, nFuture=50, M=15000,
    harvest=5, stochastic=TRUE,
    NbarSims=100,
    report=100)
})
```

```
## iter 100 of 2000
## iter 200 of 2000
```

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```
## TRUE, : Entrance probability >1. Increase M. For now, N fixed at max.
## Warning in forecast(ps, nFuture = 50, M = 15000, harvest = 5, stochastic =
## TRUE, : Entrance probability >1. Increase M. For now, N fixed at max.
## Warning in forecast(ps, nFuture = 50, M = 15000, harvest = 5, stochastic =
## TRUE, : Entrance probability >1. Increase M. For now, N fixed at max.
## Warning in forecast(ps, nFuture = 50, M = 15000, harvest = 5, stochastic =
## TRUE, : Entrance probability >1. Increase M. For now, N fixed at max.
## Warning in forecast(ps, nFuture = 50, M = 15000, harvest = 5, stochastic =
## TRUE, : Entrance probability >1. Increase M. For now, N fixed at max.
## Warning in forecast(ps, nFuture = 50, M = 15000, harvest = 5, stochastic =
## TRUE, : Entrance probability >1. Increase M. For now, N fixed at max.
## Warning in forecast(ps, nFuture = 50, M = 15000, harvest = 5, stochastic =
## TRUE, : Entrance probability >1. Increase M. For now, N fixed at max.
## Warning in forecast(ps, nFuture = 50, M = 15000, harvest = 5, stochastic =
## TRUE, : Entrance probability >1. Increase M. For now, N fixed at max.
## Warning in forecast(ps, nFuture = 50, M = 15000, harvest = 5, stochastic =
## TRUE, : Entrance probability >1. Increase M. For now, N fixed at max.
```

```
## iter 1400 of 2000
```

```
## Warning in forecast(ps, nFuture = 50, M = 15000, harvest = 5, stochastic =
## TRUE, : Entrance probability >1. Increase M. For now, N fixed at max.
```

```
## iter 1500 of 2000
## iter 1600 of 2000
## iter 1700 of 2000
## iter 1800 of 2000
## iter 1900 of 2000
## iter 2000 of 2000
```

```
## user system elapsed
## 9814.619 17.176 9869.920
```

```
save(forc.h5, file="forc-h5.gzip")
```

```
## 10 additional females per year
system.time({
  forc.h10 <- forecast(ps, nFuture=50, M=15000,
    harvest=10, stochastic=TRUE,
    NbarSims=100,
    report=100)
})
```

```
## iter 100 of 2000
## iter 200 of 2000
## iter 300 of 2000
## iter 400 of 2000
## iter 500 of 2000
## iter 600 of 2000
## iter 700 of 2000
## iter 800 of 2000
```

```
## Warning in forecast(ps, nFuture = 50, M = 15000, harvest = 10, stochastic =
## TRUE, : Entrance probability >1. Increase M. For now, N fixed at max.
```

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```
## iter 900 of 2000
## iter 1000 of 2000
## iter 1100 of 2000
## iter 1200 of 2000
## iter 1300 of 2000
```

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```
save(forc.h10, file="forc-h10.gzip")
```

```
##      iter 100 of 2000
##      iter 200 of 2000
##      iter 300 of 2000
##      iter 400 of 2000
##      iter 500 of 2000
##      iter 600 of 2000
##      iter 700 of 2000
##      iter 800 of 2000
```

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```
## iter 1400 of 2000
## iter 1500 of 2000
## iter 1600 of 2000
## iter 1700 of 2000
## iter 1800 of 2000
## iter 1900 of 2000
## iter 2000 of 2000
```

```
## user system elapsed
## 9314.060 21.546 9369.102
```

```
save(forc.h15, file="forc-h15.gzip")
```

```
## 20 additional females per year
system.time({
  forc.h20 <- forecast(ps, nFuture=50, M=15000,
    harvest=20, stochastic=TRUE,
    NbarSims=100,
    report=100)
})
```

```
## iter 100 of 2000
## iter 200 of 2000
## iter 300 of 2000
## iter 400 of 2000
## iter 500 of 2000
## iter 600 of 2000
## iter 700 of 2000
## iter 800 of 2000
```

```
## Warning in forecast(ps, nFuture = 50, M = 15000, harvest = 20, stochastic =
## TRUE, : Entrance probability >1. Increase M. For now, N fixed at max.
```

```
## Warning in forecast(ps, nFuture = 50, M = 15000, harvest = 20, stochastic =
## TRUE, : Entrance probability >1. Increase M. For now, N fixed at max.
## Warning in forecast(ps, nFuture = 50, M = 15000, harvest = 20, stochastic =
## TRUE, : Entrance probability >1. Increase M. For now, N fixed at max.
## Warning in forecast(ps, nFuture = 50, M = 15000, harvest = 20, stochastic =
## TRUE, : Entrance probability >1. Increase M. For now, N fixed at max.
## Warning in forecast(ps, nFuture = 50, M = 15000, harvest = 20, stochastic =
## TRUE, : Entrance probability >1. Increase M. For now, N fixed at max.
## Warning in forecast(ps, nFuture = 50, M = 15000, harvest = 20, stochastic =
## TRUE, : Entrance probability >1. Increase M. For now, N fixed at max.
## Warning in forecast(ps, nFuture = 50, M = 15000, harvest = 20, stochastic =
## TRUE, : Entrance probability >1. Increase M. For now, N fixed at max.
## Warning in forecast(ps, nFuture = 50, M = 15000, harvest = 20, stochastic =
## TRUE, : Entrance probability >1. Increase M. For now, N fixed at max.
## Warning in forecast(ps, nFuture = 50, M = 15000, harvest = 20, stochastic =
## TRUE, : Entrance probability >1. Increase M. For now, N fixed at max.
## Warning in forecast(ps, nFuture = 50, M = 15000, harvest = 20, stochastic =
## TRUE, : Entrance probability >1. Increase M. For now, N fixed at max.
```

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```
## Warning in forecast(ps, nFuture = 50, M = 15000, harvest = 20, stochastic =
## TRUE, : Entrance probability >1. Increase M. For now, N fixed at max.
## Warning in forecast(ps, nFuture = 50, M = 15000, harvest = 20, stochastic =
## TRUE, : Entrance probability >1. Increase M. For now, N fixed at max.
## Warning in forecast(ps, nFuture = 50, M = 15000, harvest = 20, stochastic =
## TRUE, : Entrance probability >1. Increase M. For now, N fixed at max.
## Warning in forecast(ps, nFuture = 50, M = 15000, harvest = 20, stochastic =
## TRUE, : Entrance probability >1. Increase M. For now, N fixed at max.
## Warning in forecast(ps, nFuture = 50, M = 15000, harvest = 20, stochastic =
## TRUE, : Entrance probability >1. Increase M. For now, N fixed at max.

##   iter 1400 of 2000
##   iter 1500 of 2000
##   iter 1600 of 2000
##   iter 1700 of 2000
##   iter 1800 of 2000
##   iter 1900 of 2000
##   iter 2000 of 2000

##      user      system elapsed
## 8589.179      15.258 8635.601
```

```
save(forc.h20, file="forc-h20.gzip")
```

Compare forecasted abundance

Now we'll look at the distribution of projected population sizes under each of the management alternatives (*status quo*, +5, +10, +15, or +20 additional females harvested) and ask where our median parameter estimates for female abundance in 2023 and 2024 fall in those distributions.

```
load("forc-h0.gzip")
load("forc-h5.gzip")
load("forc-h10.gzip")
load("forc-h15.gzip")
load("forc-h20.gzip")

#Calculate median population size for each draw from the posterior
# 2000 total draws, each simulated 100 times over a period of 50 years
dim(forc.h0$Nbar)
```

```
## [1] 51 100 2000
```

```
dim(forc.h5$Nbar)
```

```
## [1] 51 100 2000
```

```
dim(forc.h10$Nbar)
```

```
## [1] 51 100 2000
```

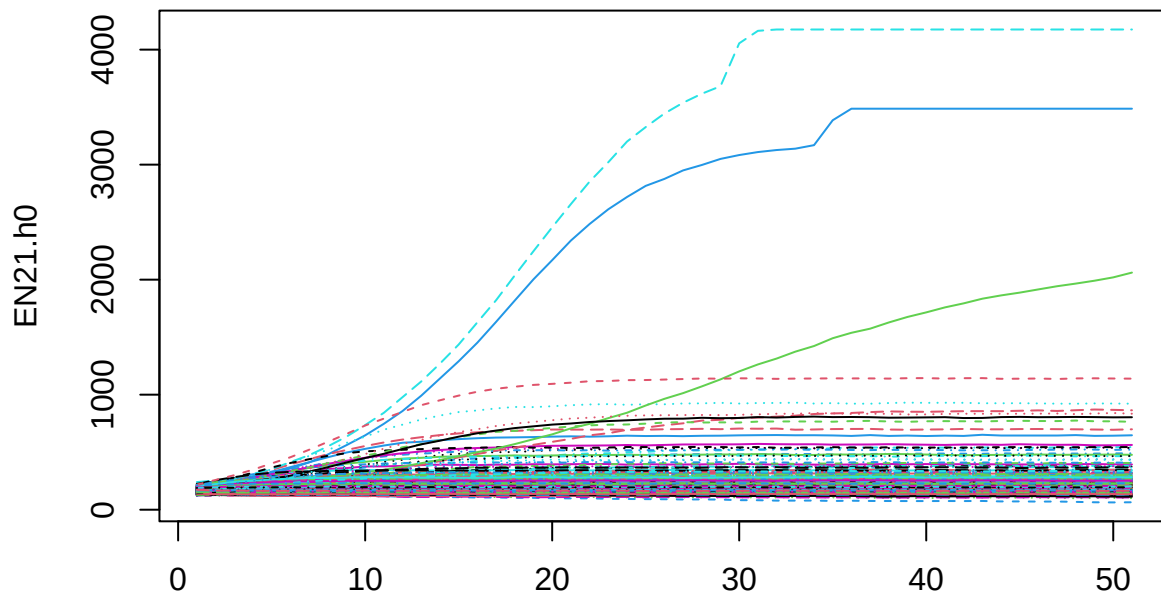
```
dim(forc.h15$Nbar)
```

```
## [1] 51 100 2000
```

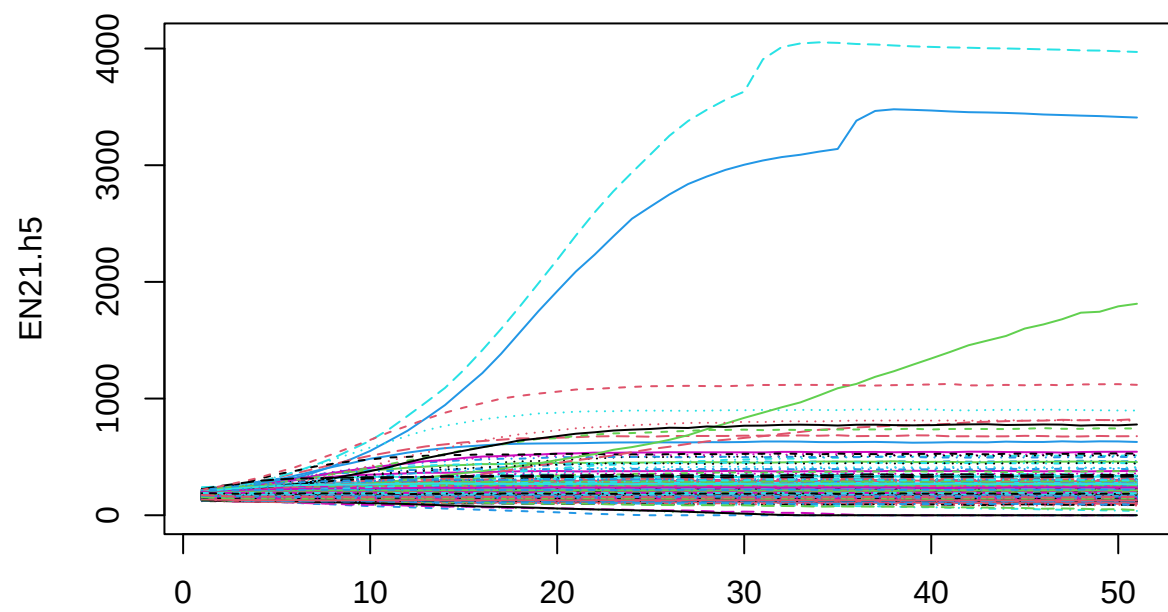
```
dim(forc.h20$Nbar)
```

```
## [1] 51 100 2000
```

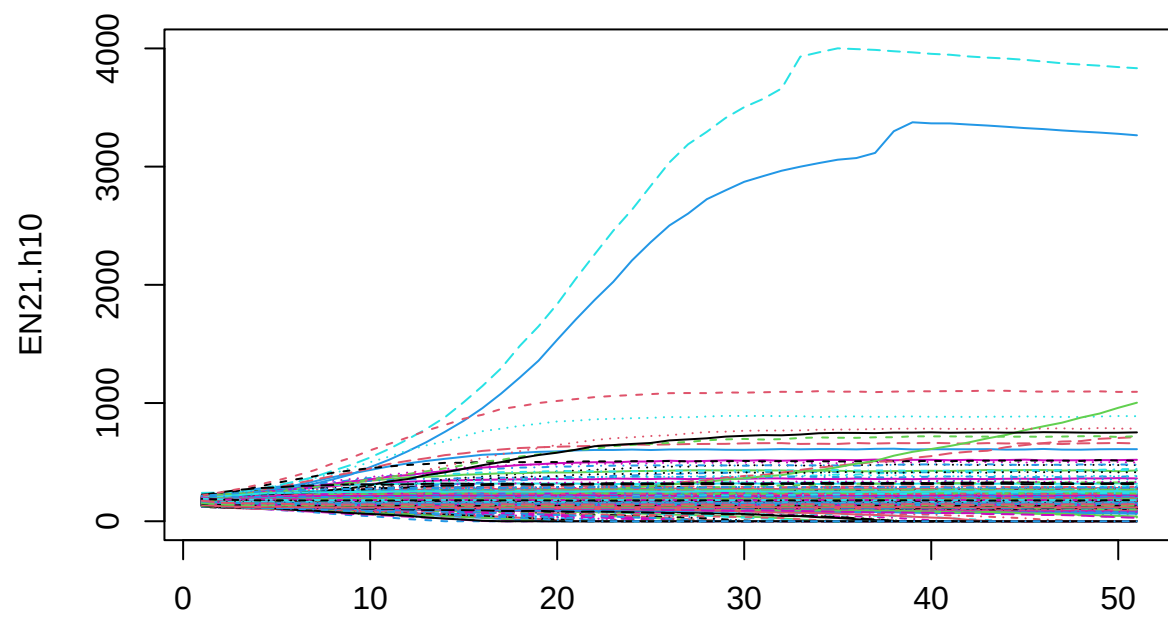
```
# ---  
#From Hooker et al. (2020) forecast.R script  
EN21.h0 <- apply(forc.h0$Nbar, c(1, 3), median)  
EN21.h5 <- apply(forc.h5$Nbar, c(1, 3), median)  
EN21.h10 <- apply(forc.h10$Nbar, c(1, 3), median)  
EN21.h15 <- apply(forc.h15$Nbar, c(1, 3), median)  
EN21.h20 <- apply(forc.h20$Nbar, c(1, 3), median)  
  
matplot(EN21.h0, type="l")
```



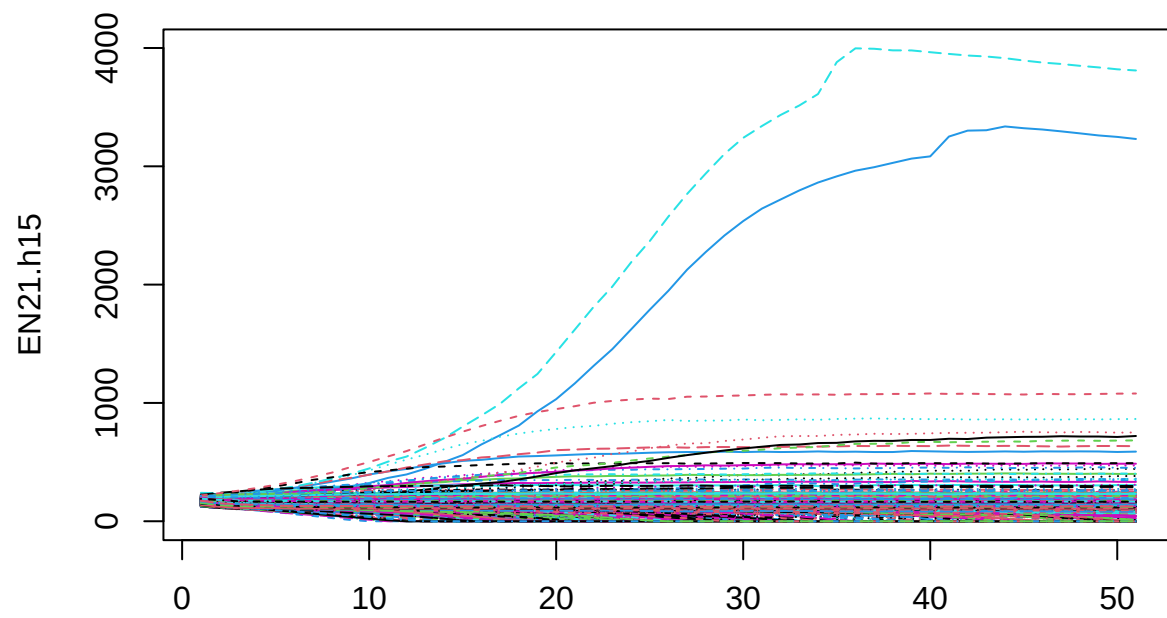
```
matplot(EN21.h5, type="l")
```



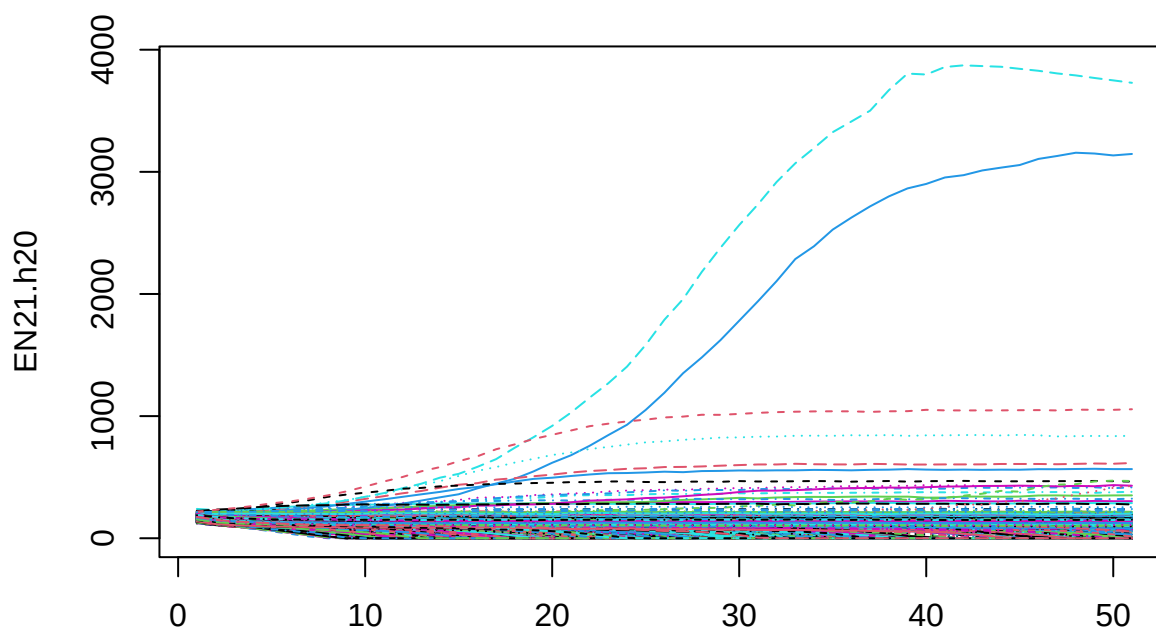
```
matplot(EN21.h10, type="l")
```



```
matplot(EN21.h15, type="l")
```



```
matplot(EN21.h20, type="l")
```

```
quantile(EN21.h0[51,], c(0.025, 0.5, 0.975))
```

```
##      2.5%      50%      97.5%
## 134.4875 171.5000 306.0250
```

```
quantile(EN21.h5[51,], c(0.025, 0.5, 0.975))
```

```
##      2.5%      50%      97.5%
## 122.5000 162.0000 284.0125
```

```
quantile(EN21.h10[51,], c(0.025, 0.5, 0.975))
```

```
##      2.5%      50%      97.5%
##  98.9375 149.7500 257.0500
```

```
quantile(EN21.h15[51,], c(0.025, 0.5, 0.975))
```

```
##      2.5%      50%      97.5%
##   0.0000 134.0000 220.5375
```

```
quantile(EN21.h20[51,], c(0.025, 0.5, 0.975))
```

```
##      2.5%      50%      97.5%
##      0.000 103.000 183.025
```

```
## Years
fyrs <- 2017:(2017+50)
# ---
```

Visual comparisons

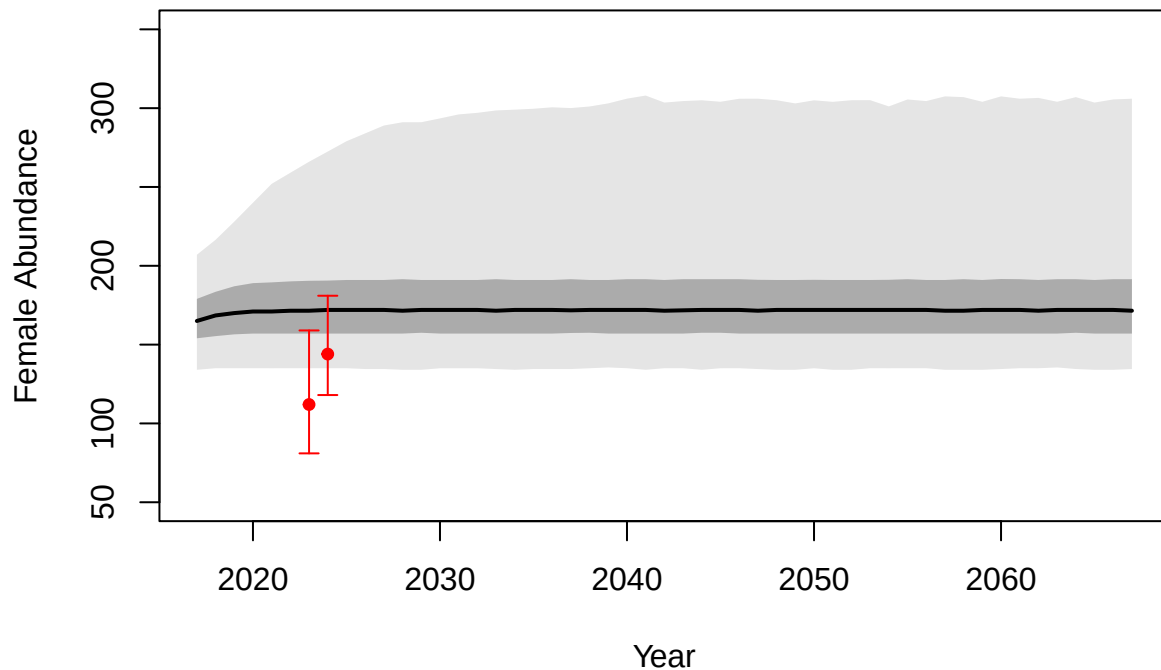
Now plot the full time series with our abundance estimates included. We focus on the v1 model, with a single recruitment rate parameter shared by males and females, as the estimates of γ are more reliable under this model.

```
library(scales)

#Load point estimates
load("../scr/samples-bothSexes_v1.gzip")
out <- coda::as.mcmc.list(samples)
Nf23 <- unlist(out[, "Nf[1]"])
Nf24 <- unlist(out[, "Nf[2]"])

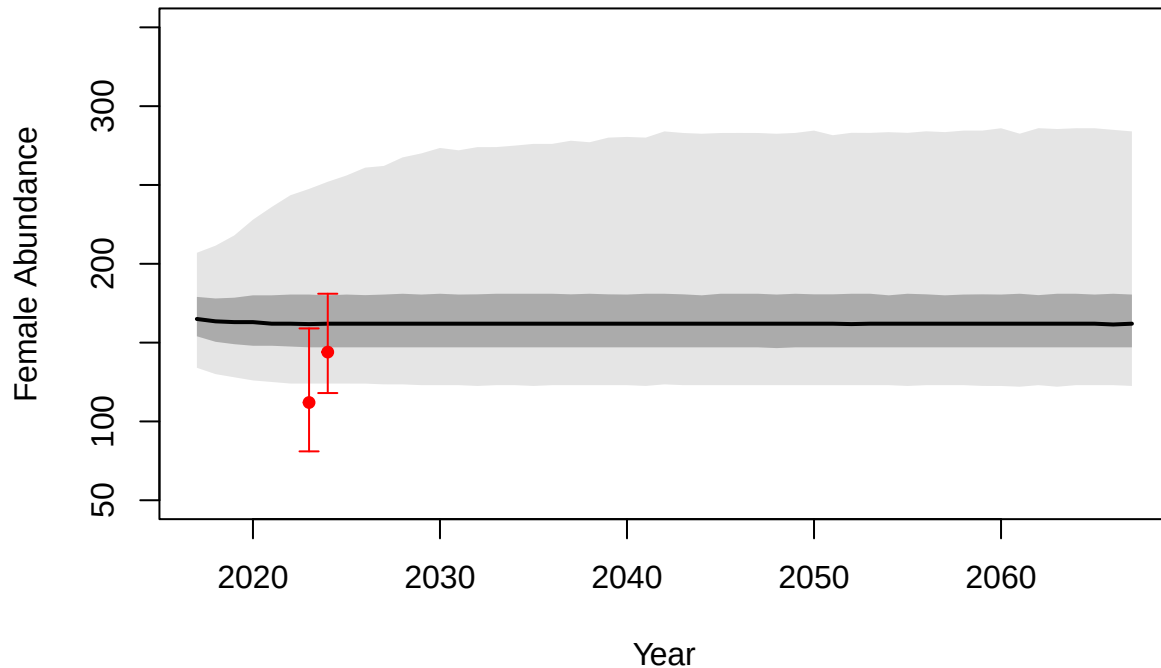
#First for status quo
plot(fyrs, apply(EN21.h0, 1, median), type = "l",
     ylim = c(50, 350), lwd = 2, col = "black",
     xlab = "Year", ylab = "Female Abundance", main = "Status Quo")
HPD50.h0 <- apply(EN21.h0, 1, quantile, prob = c(0.25, 0.75))
HPD95.h0 <- apply(EN21.h0, 1, quantile, prob = c(0.025, 0.975))
polygon(x=c(fyrs, fyrs[length(fyrs):1]),
        y=c(HPD50.h0[1,], HPD50.h0[2,][ncol(HPD50.h0):1]),
        col = alpha("black", 0.25), border = NA)
polygon(x=c(fyrs, fyrs[length(fyrs):1]),
        y=c(HPD95.h0[1,], HPD95.h0[2,][ncol(HPD95.h0):1]),
        col = alpha("black", 0.1), border = NA)
#Add points for estimates (and HPDI)
points(x = 2023, y = median(Nf23), pch = 19, cex = 0.75, col = "red")
arrows(x0 = 2023, x1 = 2023,
       y0 = quantile(Nf23, 0.025), y1 = quantile(Nf23, 0.975),
       length = 0.05, angle = 90, code = 3, col = "red")
points(x = 2024, y = median(Nf24), pch = 19, cex = 0.75, col = "red")
arrows(x0 = 2024, x1 = 2024,
       y0 = quantile(Nf24, 0.025), y1 = quantile(Nf24, 0.975),
       length = 0.05, angle = 90, code = 3, col = "red")
```

Status Quo



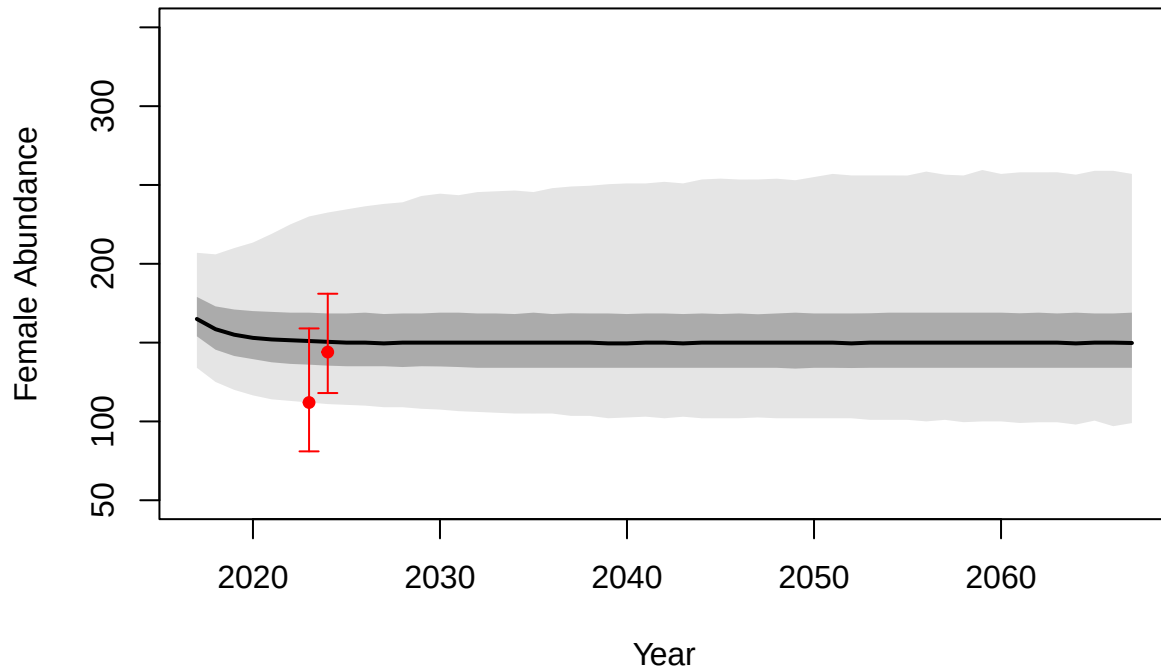
```
#Then for 5 additional females per year
plot(fyrs, apply(EN21.h5, 1, median), type = "l",
     ylim = c(50,350), lwd = 2, col = "black",
     xlab = "Year", ylab = "Female Abundance", main = "+5 Females")
HPD50.h5 <- apply(EN21.h5, 1, quantile, prob = c(0.25, 0.75))
HPD95.h5 <- apply(EN21.h5, 1, quantile, prob = c(0.025, 0.975))
polygon(x=c(fyrs, fyrs[length(fyrs):1]),
       y=c(HPD50.h5[1,], HPD50.h5[2,][ncol(HPD50.h5):1]),
       col = alpha("black", 0.25), border = NA)
polygon(x=c(fyrs, fyrs[length(fyrs):1]),
       y=c(HPD95.h5[1,], HPD95.h5[2,][ncol(HPD95.h5):1]),
       col = alpha("black", 0.1), border = NA)
#Add points for estimates (and HPDI)
points(x = 2023, y = median(Nf23), pch = 19, cex = 0.75, col = "red")
arrows(x0 = 2023, x1 = 2023,
      y0 = quantile(Nf23, 0.025), y1 = quantile(Nf23, 0.975),
      length = 0.05, angle = 90, code = 3, col = "red")
points(x = 2024, y = median(Nf24), pch = 19, cex = 0.75, col = "red")
arrows(x0 = 2024, x1 = 2024,
      y0 = quantile(Nf24, 0.025), y1 = quantile(Nf24, 0.975),
      length = 0.05, angle = 90, code = 3, col = "red")
```

+5 Females



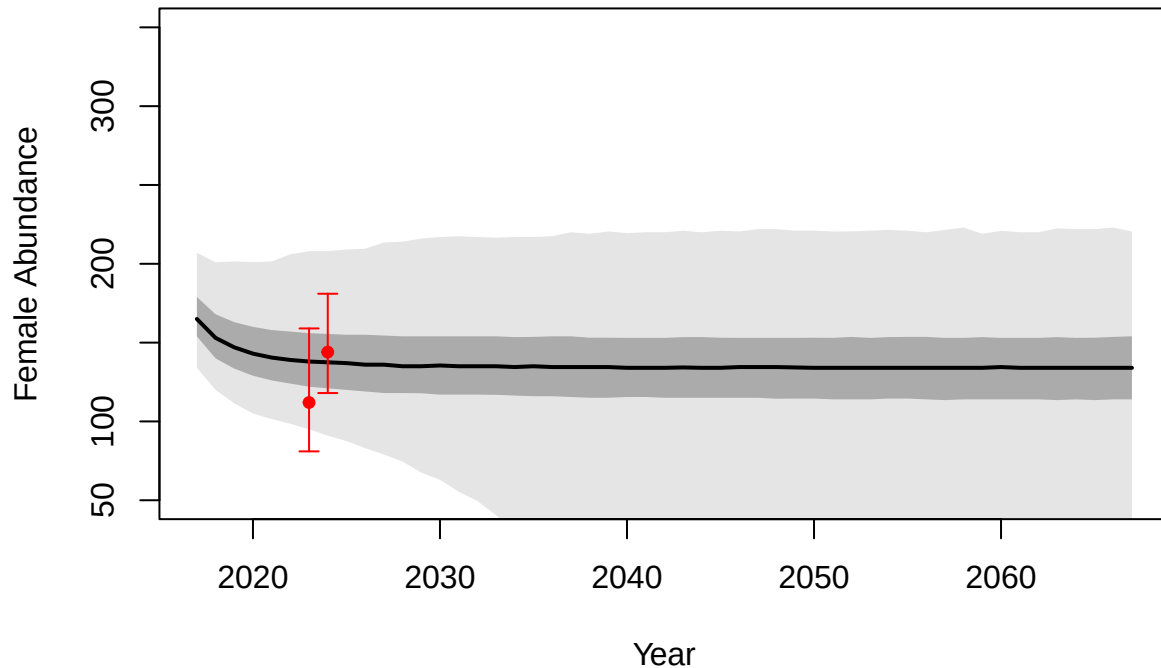
```
#Then for 10 additional females per year
plot(fyrs, apply(EN21.h10, 1, median), type = "l",
     ylim = c(50,350), lwd = 2, col = "black",
     xlab = "Year", ylab = "Female Abundance", main = "+10 Females")
HPD50.h10 <- apply(EN21.h10, 1, quantile, prob = c(0.25, 0.75))
HPD95.h10 <- apply(EN21.h10, 1, quantile, prob = c(0.025, 0.975))
polygon(x=c(fyrs, fyrs[length(fyrs):1]),
       y=c(HPD50.h10[1,], HPD50.h10[2,][ncol(HPD50.h10):1]),
       col = alpha("black", 0.25), border = NA)
polygon(x=c(fyrs, fyrs[length(fyrs):1]),
       y=c(HPD95.h10[1,], HPD95.h10[2,][ncol(HPD95.h10):1]),
       col = alpha("black", 0.1), border = NA)
#Add points for estimates (and HPDI)
points(x = 2023, y = median(Nf23), pch = 19, cex = 0.75, col = "red")
arrows(x0 = 2023, x1 = 2023,
       y0 = quantile(Nf23, 0.025), y1 = quantile(Nf23, 0.975),
       length = 0.05, angle = 90, code = 3, col = "red")
points(x = 2024, y = median(Nf24), pch = 19, cex = 0.75, col = "red")
arrows(x0 = 2024, x1 = 2024,
       y0 = quantile(Nf24, 0.025), y1 = quantile(Nf24, 0.975),
       length = 0.05, angle = 90, code = 3, col = "red")
```

+10 Females



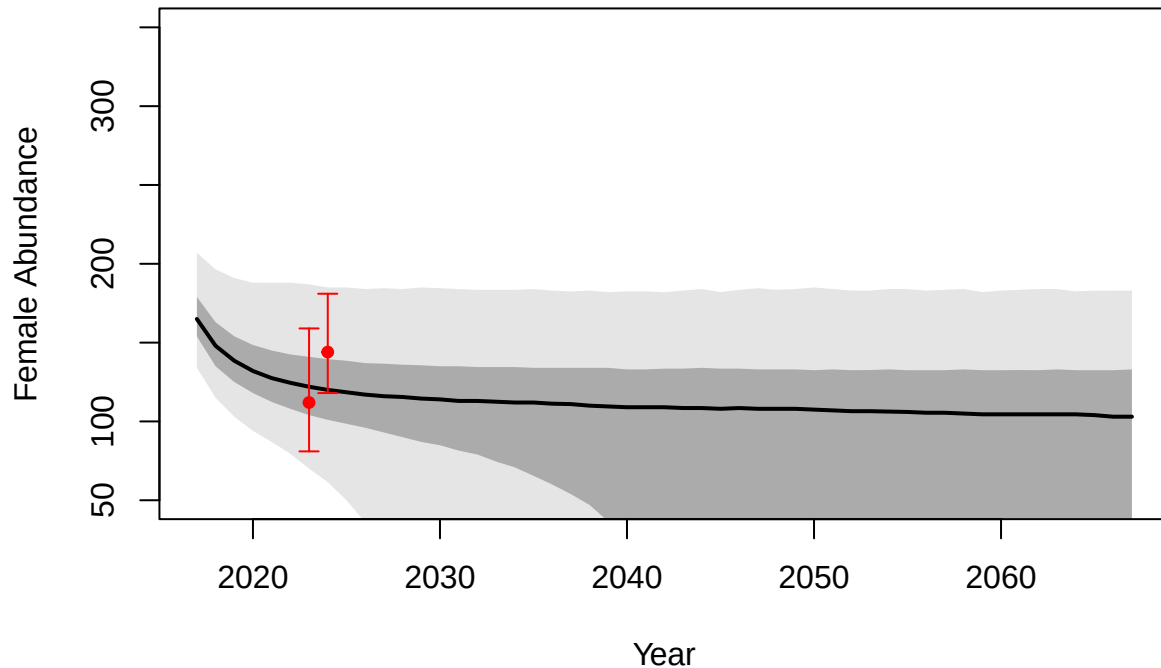
```
#Then for 15 additional females per year
plot(fyrs, apply(EN21.h15, 1, median), type = "l",
     ylim = c(50,350), lwd = 2, col = "black",
     xlab = "Year", ylab = "Female Abundance", main = "+15 Females")
HPD50.h15 <- apply(EN21.h15, 1, quantile, prob = c(0.25, 0.75))
HPD95.h15 <- apply(EN21.h15, 1, quantile, prob = c(0.025, 0.975))
polygon(x=c(fyrs, fyrs[length(fyrs):1]),
        y=c(HPD50.h15[1,], HPD50.h15[2,][ncol(HPD50.h15):1]),
        col = alpha("black", 0.25), border = NA)
polygon(x=c(fyrs, fyrs[length(fyrs):1]),
        y=c(HPD95.h15[1,], HPD95.h15[2,][ncol(HPD95.h15):1]),
        col = alpha("black", 0.1), border = NA)
#Add points for estimates (and HPDI)
points(x = 2023, y = median(Nf23), pch = 19, cex = 0.75, col = "red")
arrows(x0 = 2023, x1 = 2023,
       y0 = quantile(Nf23, 0.025), y1 = quantile(Nf23, 0.975),
       length = 0.05, angle = 90, code = 3, col = "red")
points(x = 2024, y = median(Nf24), pch = 19, cex = 0.75, col = "red")
arrows(x0 = 2024, x1 = 2024,
       y0 = quantile(Nf24, 0.025), y1 = quantile(Nf24, 0.975),
       length = 0.05, angle = 90, code = 3, col = "red")
```

+15 Females



```
#Then for 20 additional females per year
plot(fyrs, apply(EN21.h20, 1, median), type = "l",
     ylim = c(50,350), lwd = 2, col = "black",
     xlab = "Year", ylab = "Female Abundance", main = "+20 Females")
HPD50.h20 <- apply(EN21.h20, 1, quantile, prob = c(0.25, 0.75))
HPD95.h20 <- apply(EN21.h20, 1, quantile, prob = c(0.025, 0.975))
polygon(x=c(fyrs, fyrs[length(fyrs):1]),
       y=c(HPD50.h20[1,], HPD50.h20[2,][ncol(HPD50.h20):1]),
       col = alpha("black", 0.25), border = NA)
polygon(x=c(fyrs, fyrs[length(fyrs):1]),
       y=c(HPD95.h20[1,], HPD95.h20[2,][ncol(HPD95.h20):1]),
       col = alpha("black", 0.1), border = NA)
#Add points for estimates (and HPDI)
points(x = 2023, y = median(Nf23), pch = 19, cex = 0.75, col = "red")
arrows(x0 = 2023, x1 = 2023,
       y0 = quantile(Nf23, 0.025), y1 = quantile(Nf23, 0.975),
       length = 0.05, angle = 90, code = 3, col = "red")
points(x = 2024, y = median(Nf24), pch = 19, cex = 0.75, col = "red")
arrows(x0 = 2024, x1 = 2024,
       y0 = quantile(Nf24, 0.025), y1 = quantile(Nf24, 0.975),
       length = 0.05, angle = 90, code = 3, col = "red")
```

+20 Females



Zooming in on the first part of the forecast...

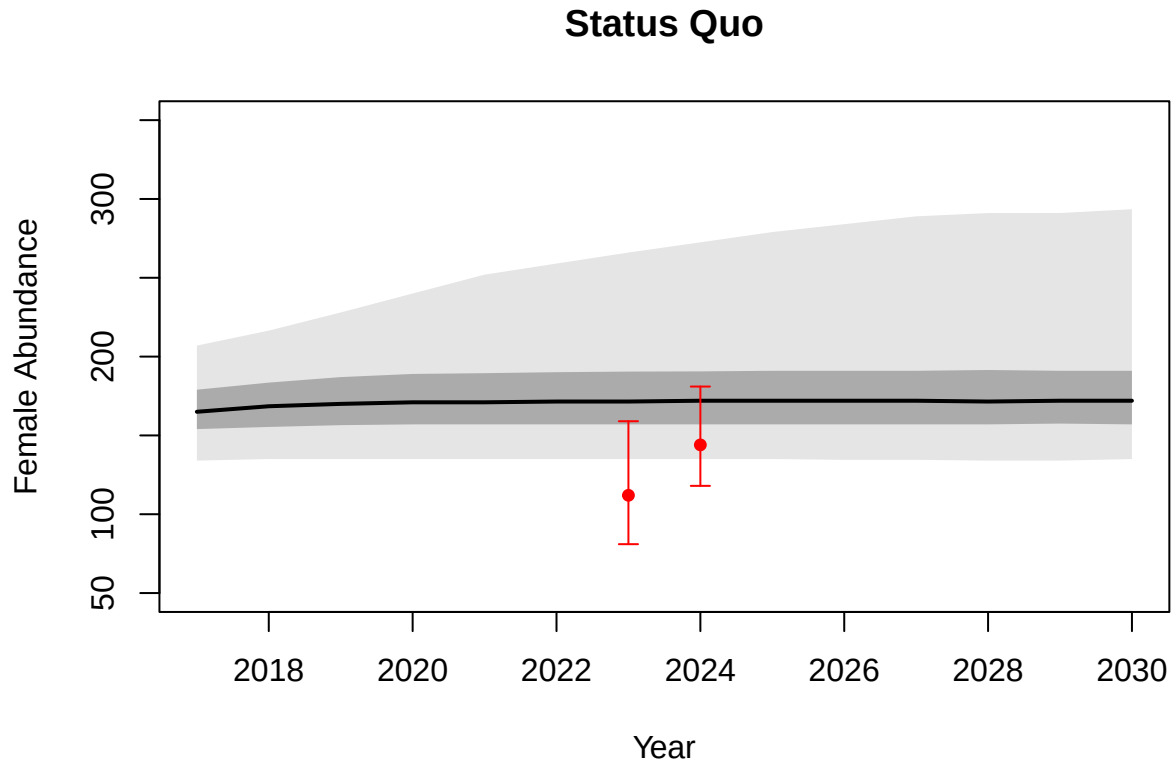
```
max.yr <- which(fyrs == 2030)
fyrs <- fyrs[1:max.yr]
EN21.h0 <- EN21.h0[1:max.yr,]
EN21.h5 <- EN21.h5[1:max.yr,]
EN21.h10 <- EN21.h10[1:max.yr,]
EN21.h15 <- EN21.h15[1:max.yr,]
EN21.h20 <- EN21.h20[1:max.yr,]

#First for status quo
plot(fyrs, apply(EN21.h0, 1, median), type = "l",
     ylim = c(50,350), lwd = 2, col = "black",
     xlab = "Year", ylab = "Female Abundance", main = "Status Quo")
HPD50.h0 <- apply(EN21.h0, 1, quantile, prob = c(0.25, 0.75))
HPD95.h0 <- apply(EN21.h0, 1, quantile, prob = c(0.025, 0.975))
polygon(x=c(fyrs, fyrs[length(fyrs):1]),
       y=c(HPD50.h0[1,], HPD50.h0[2,][ncol(HPD50.h0):1]),
       col = alpha("black", 0.25), border = NA)
polygon(x=c(fyrs, fyrs[length(fyrs):1]),
       y=c(HPD95.h0[1,], HPD95.h0[2,][ncol(HPD95.h0):1]),
       col = alpha("black", 0.1), border = NA)
#Add points for estimates (and HPDI)
points(x = 2023, y = median(Nf23), pch = 19, cex = 0.75, col = "red")
arrows(x0 = 2023, x1 = 2023,
       y0 = quantile(Nf23, 0.025), y1 = quantile(Nf23, 0.975),
```

```

length = 0.05, angle = 90, code = 3, col = "red")
points(x = 2024, y = median(Nf24), pch = 19, cex = 0.75, col = "red")
arrows(x0 = 2024, x1 = 2024,
       y0 = quantile(Nf24, 0.025), y1 = quantile(Nf24, 0.975),
       length = 0.05, angle = 90, code = 3, col = "red")

```



```

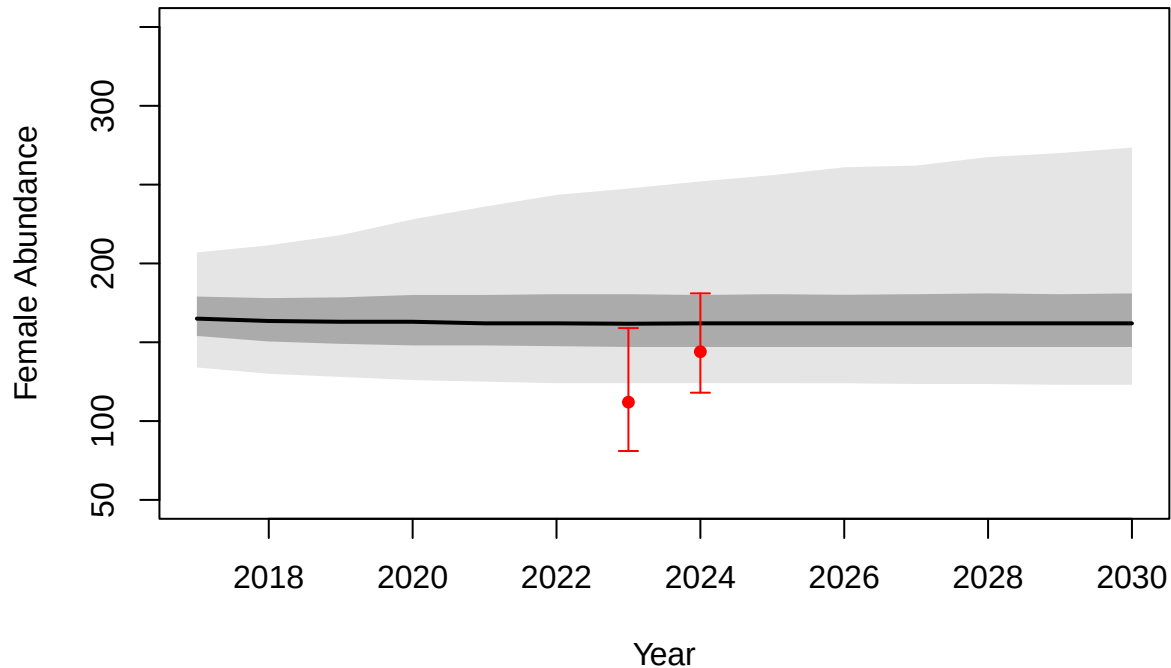
#Then for 5 additional females per year
plot(fyrs, apply(EN21.h5, 1, median), type = "l",
     ylim = c(50,350), lwd = 2, col = "black",
     xlab = "Year", ylab = "Female Abundance", main = "+5 Females")
HPD50.h5 <- apply(EN21.h5, 1, quantile, prob = c(0.25, 0.75))
HPD95.h5 <- apply(EN21.h5, 1, quantile, prob = c(0.025, 0.975))
polygon(x=c(fyrs, fyrs[length(fyrs):1]),
       y=c(HPD50.h5[1,], HPD50.h5[2,][ncol(HPD50.h5):1]),
       col = alpha("black", 0.25), border = NA)
polygon(x=c(fyrs, fyrs[length(fyrs):1]),
       y=c(HPD95.h5[1,], HPD95.h5[2,][ncol(HPD95.h5):1]),
       col = alpha("black", 0.1), border = NA)
#Add points for estimates (and HPDI)
points(x = 2023, y = median(Nf23), pch = 19, cex = 0.75, col = "red")
arrows(x0 = 2023, x1 = 2023,
       y0 = quantile(Nf23, 0.025), y1 = quantile(Nf23, 0.975),
       length = 0.05, angle = 90, code = 3, col = "red")
points(x = 2024, y = median(Nf24), pch = 19, cex = 0.75, col = "red")
arrows(x0 = 2024, x1 = 2024,

```



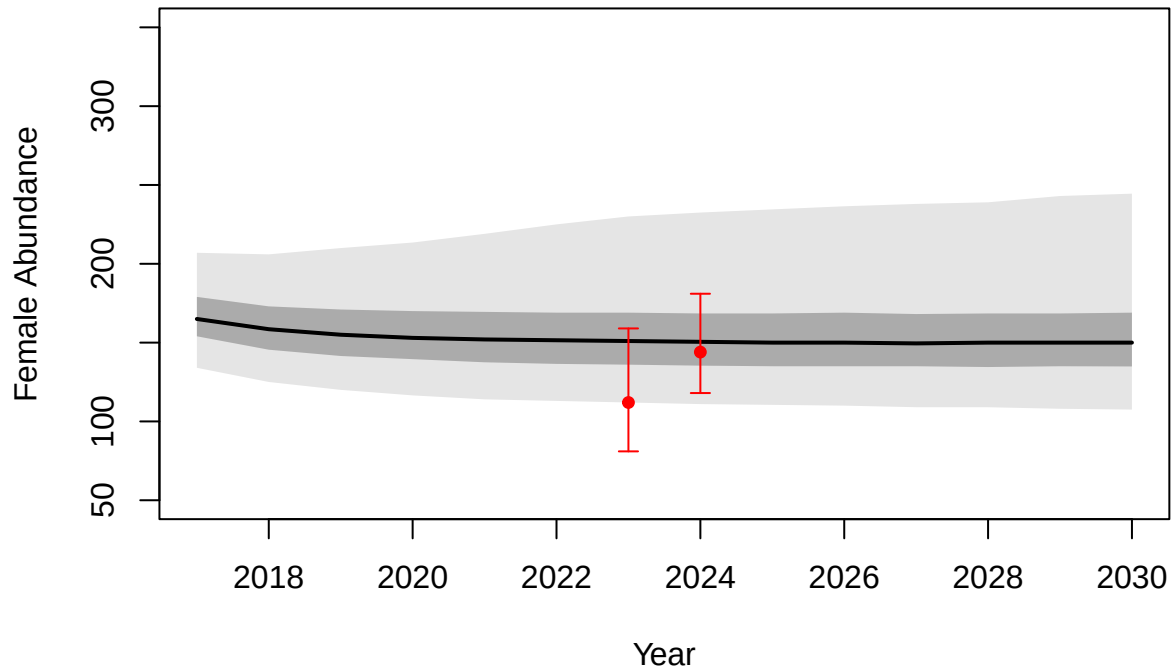
```
y0 = quantile(Nf24, 0.025), y1 = quantile(Nf24, 0.975),
length = 0.05, angle = 90, code = 3, col = "red")
```

+5 Females



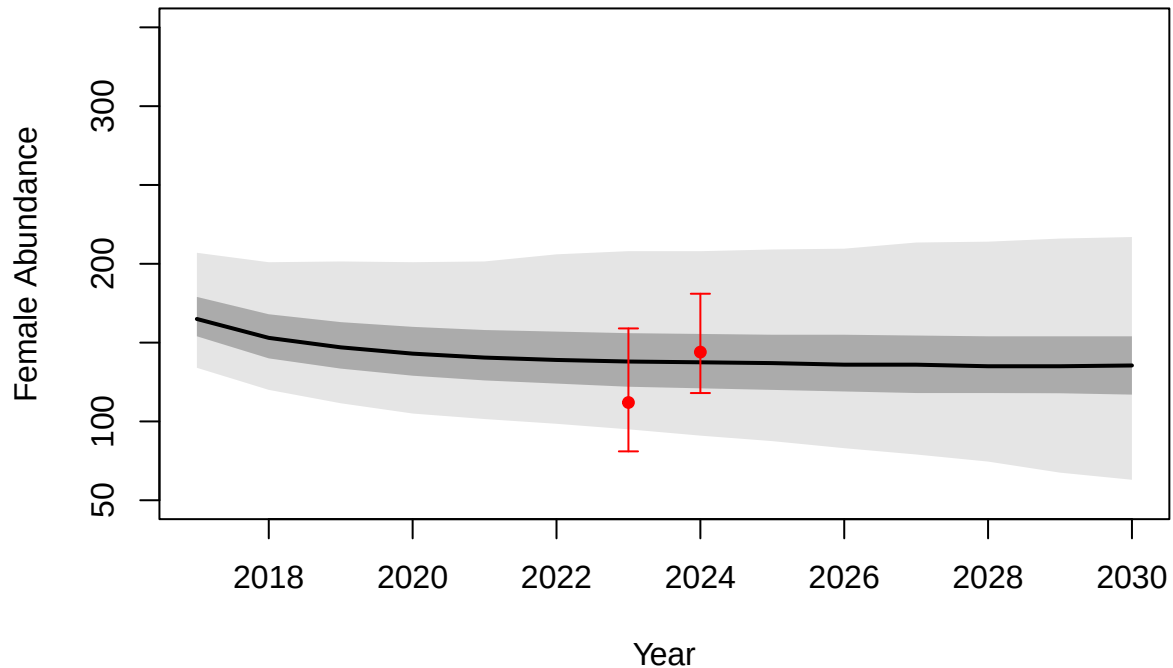
```
#Then for 10 additional females per year
plot(fyrs, apply(EN21.h10, 1, median), type = "l",
     ylim = c(50,350), lwd = 2, col = "black",
     xlab = "Year", ylab = "Female Abundance", main = "+10 Females")
HPD50.h10 <- apply(EN21.h10, 1, quantile, prob = c(0.25, 0.75))
HPD95.h10 <- apply(EN21.h10, 1, quantile, prob = c(0.025, 0.975))
polygon(x=c(fyrs, fyrs[length(fyrs):1]),
       y=c(HPD50.h10[1,], HPD50.h10[2,][ncol(HPD50.h10):1]),
       col = alpha("black", 0.25), border = NA)
polygon(x=c(fyrs, fyrs[length(fyrs):1]),
       y=c(HPD95.h10[1,], HPD95.h10[2,][ncol(HPD95.h10):1]),
       col = alpha("black", 0.1), border = NA)
#Add points for estimates (and HPDI)
points(x = 2023, y = median(Nf23), pch = 19, cex = 0.75, col = "red")
arrows(x0 = 2023, x1 = 2023,
      y0 = quantile(Nf23, 0.025), y1 = quantile(Nf23, 0.975),
      length = 0.05, angle = 90, code = 3, col = "red")
points(x = 2024, y = median(Nf24), pch = 19, cex = 0.75, col = "red")
arrows(x0 = 2024, x1 = 2024,
      y0 = quantile(Nf24, 0.025), y1 = quantile(Nf24, 0.975),
      length = 0.05, angle = 90, code = 3, col = "red")
```

+10 Females



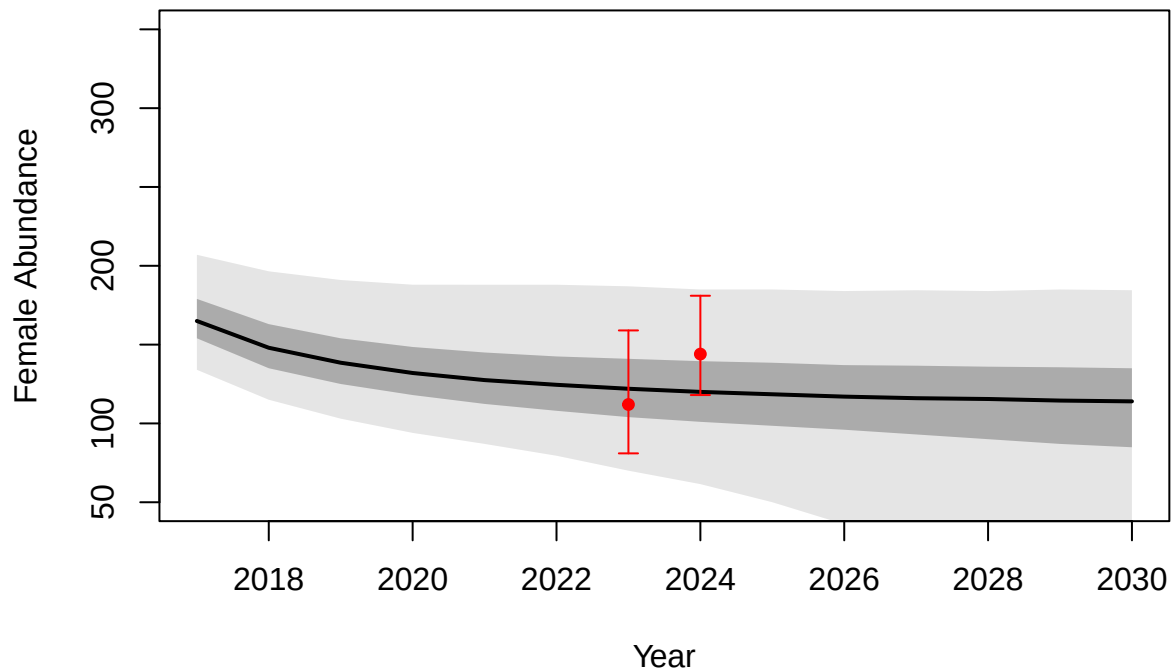
```
#Then for 15 additional females per year
plot(fyrs, apply(EN21.h15, 1, median), type = "l",
     ylim = c(50,350), lwd = 2, col = "black",
     xlab = "Year", ylab = "Female Abundance", main = "+15 Females")
HPD50.h15 <- apply(EN21.h15, 1, quantile, prob = c(0.25, 0.75))
HPD95.h15 <- apply(EN21.h15, 1, quantile, prob = c(0.025, 0.975))
polygon(x=c(fyrs, fyrs[length(fyrs):1]),
       y=c(HPD50.h15[1,], HPD50.h15[2,][ncol(HPD50.h15):1]),
       col = alpha("black", 0.25), border = NA)
polygon(x=c(fyrs, fyrs[length(fyrs):1]),
       y=c(HPD95.h15[1,], HPD95.h15[2,][ncol(HPD95.h15):1]),
       col = alpha("black", 0.1), border = NA)
#Add points for estimates (and HPDI)
points(x = 2023, y = median(Nf23), pch = 19, cex = 0.75, col = "red")
arrows(x0 = 2023, x1 = 2023,
       y0 = quantile(Nf23, 0.025), y1 = quantile(Nf23, 0.975),
       length = 0.05, angle = 90, code = 3, col = "red")
points(x = 2024, y = median(Nf24), pch = 19, cex = 0.75, col = "red")
arrows(x0 = 2024, x1 = 2024,
       y0 = quantile(Nf24, 0.025), y1 = quantile(Nf24, 0.975),
       length = 0.05, angle = 90, code = 3, col = "red")
```

+15 Females



```
#Then for 20 additional females per year
plot(fyrs, apply(EN21.h20, 1, median), type = "l",
     ylim = c(50,350), lwd = 2, col = "black",
     xlab = "Year", ylab = "Female Abundance", main = "+20 Females")
HPD50.h20 <- apply(EN21.h20, 1, quantile, prob = c(0.25, 0.75))
HPD95.h20 <- apply(EN21.h20, 1, quantile, prob = c(0.025, 0.975))
polygon(x=c(fyrs, fyrs[length(fyrs):1]),
       y=c(HPD50.h20[1,], HPD50.h20[2,][ncol(HPD50.h20):1]),
       col = alpha("black", 0.25), border = NA)
polygon(x=c(fyrs, fyrs[length(fyrs):1]),
       y=c(HPD95.h20[1,], HPD95.h20[2,][ncol(HPD95.h20):1]),
       col = alpha("black", 0.1), border = NA)
#Add points for estimates (and HPDI)
points(x = 2023, y = median(Nf23), pch = 19, cex = 0.75, col = "red")
arrows(x0 = 2023, x1 = 2023,
       y0 = quantile(Nf23, 0.025), y1 = quantile(Nf23, 0.975),
       length = 0.05, angle = 90, code = 3, col = "red")
points(x = 2024, y = median(Nf24), pch = 19, cex = 0.75, col = "red")
arrows(x0 = 2024, x1 = 2024,
       y0 = quantile(Nf24, 0.025), y1 = quantile(Nf24, 0.975),
       length = 0.05, angle = 90, code = 3, col = "red")
```

+20 Females



Quantify location in distribution

Where do our point estimates fall in the distribution of simulations? In other words, what percentile is associated with the medians of our posterior distributions for abundance?

```
#Create ecdf for each year and scenario
#Status quo
ecdf.h0.2023 <- ecdf(EN21.h0[which(fyrs == 2023),])
ecdf.h0.2023(median(Nf23))
```

```
## [1] 0
```

```
ecdf.h0.2024 <- ecdf(EN21.h0[which(fyrs == 2024),])
ecdf.h0.2024(median(Nf24))
```

```
## [1] 0.099
```

```
#removing +5 females
ecdf.h5.2023 <- ecdf(EN21.h5[which(fyrs == 2023),])
ecdf.h5.2023(median(Nf23))
```

```
## [1] 0.0035
```

```
ecdf.h5.2024 <- ecdf(EN21.h5[which(fyrs == 2024),])
ecdf.h5.2024(median(Nf24))
```

```
## [1] 0.21
```

```
#removing +10 females
ecdf.h10.2023 <- ecdf(EN21.h10[which(fyrs == 2023),])
ecdf.h10.2023(median(Nf23))
```

```
## [1] 0.0255
```

```
ecdf.h10.2024 <- ecdf(EN21.h10[which(fyrs == 2024),])
ecdf.h10.2024(median(Nf24))
```

```
## [1] 0.396
```

```
#removing +15 females
ecdf.h15.2023 <- ecdf(EN21.h15[which(fyrs == 2023),])
ecdf.h15.2023(median(Nf23))
```

```
## [1] 0.1315
```

```
ecdf.h15.2024 <- ecdf(EN21.h15[which(fyrs == 2024),])
ecdf.h15.2024(median(Nf24))
```

```
## [1] 0.613
```

```
#removing +20 females
ecdf.h20.2023 <- ecdf(EN21.h20[which(fyrs == 2023),])
ecdf.h20.2023(median(Nf23))
```

```
## [1] 0.3525
```

```
ecdf.h20.2024 <- ecdf(EN21.h20[which(fyrs == 2024),])
ecdf.h20.2024(median(Nf24))
```

```
## [1] 0.7925
```

Quantifying “fit” of the scenarios

Can we distinguish between the “fit” of the scenarios simulated in Hooker et al. (2020) to our abundance estimates from 2023 and 2024? There is substantial overlap in the predictions from the model with more limited harvest rates, partially due to the assumption of Poisson-distributed annual harvest (*i.e.*, with $h = 5$, we are not removing exactly 5 bears from the population each year). Notably, annual known mortality numbers since the Hooker et al. (2020) study were most similar to the *status quo* or 5 additional females removed scenarios (see above). However, we are still interested in assessing whether the different scenarios generate distinguishable predictions for abundance.

To assess the fit of our estimates (*i.e.*, posterior distributions of female abundance in 2023 and 2024) to the management scenarios from Hooker et al. (2020), we will calculate Kullback-Leibler divergence (D_{kl})

between the distributions of simulated abundance under each scenario and the posterior distribution of estimated abundance for each of the two study years. We'll use the `kld_est_nn()` function from the `kldest` R package (Hartung 2024) to estimate KL divergence between our forecasts and posteriors with a nearest-neighbor approach.

```
#First collect the distributions of abundance from the forecasts
# combine these into one data frame from each of the scenarios
# these will be the distributions that we compare to posteriors
h0df <- data.frame(n23 = EN21.h0[which(fyrs == 2023),],
                  n24 = EN21.h0[which(fyrs == 2024),])
h5df <- data.frame(n23 = EN21.h5[which(fyrs == 2023),],
                  n24 = EN21.h5[which(fyrs == 2024),])
h10df <- data.frame(n23 = EN21.h10[which(fyrs == 2023),],
                   n24 = EN21.h10[which(fyrs == 2024),])
h15df <- data.frame(n23 = EN21.h15[which(fyrs == 2023),],
                   n24 = EN21.h15[which(fyrs == 2024),])
h20df <- data.frame(n23 = EN21.h20[which(fyrs == 2023),],
                   n24 = EN21.h20[which(fyrs == 2024),])

#Now combine posteriors in the same fashion
# this is the "true" distribution we're seeking to approximate
load("../scr/samples-bothSexes_v1.gzip")
out <- coda::as.mcmc.list(samples)

est.df <- data.frame(n23 = unlist(out[, "Nf[1]"]),
                    n24 = unlist(out[, "Nf[2]"]))

#Now calculate divergence for each scenario, using 50 nearest neighbors...
library(kldest)

kld_est_nn(X=est.df, Y=h0df, k=50) #status quo

## [1] 2.904076

kld_est_nn(X=est.df, Y=h5df, k=50) #+5 females

## [1] 2.606116

kld_est_nn(X=est.df, Y=h10df, k=50) #+10 females

## [1] 2.465135

kld_est_nn(X=est.df, Y=h15df, k=50) #+15 females

## [1] 2.456261

kld_est_nn(X=est.df, Y=h20df, k=50) #+20 females

## [1] 2.513953
```